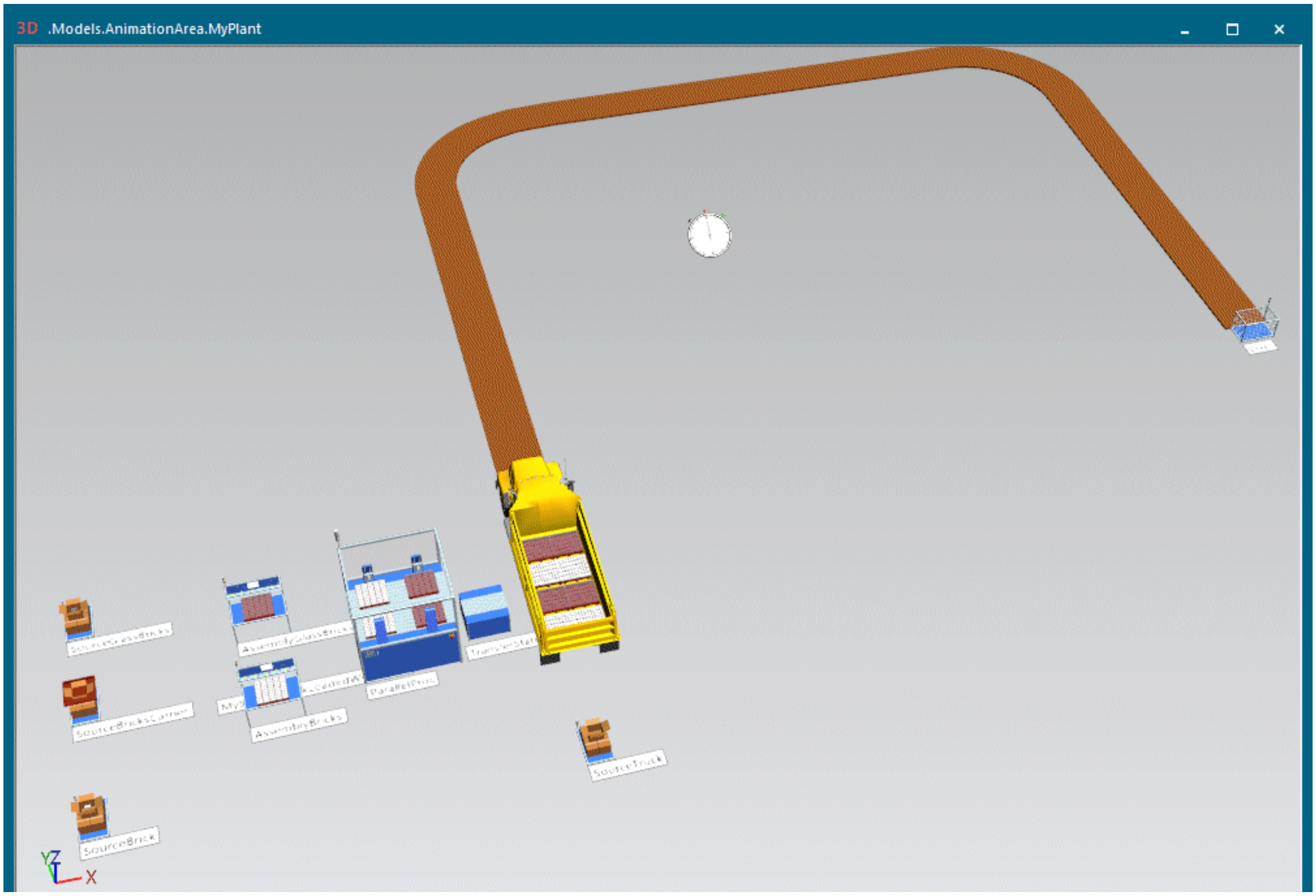


Back to Model a Complex Receiving Department

Working with Animation Paths

Work with Animation Paths

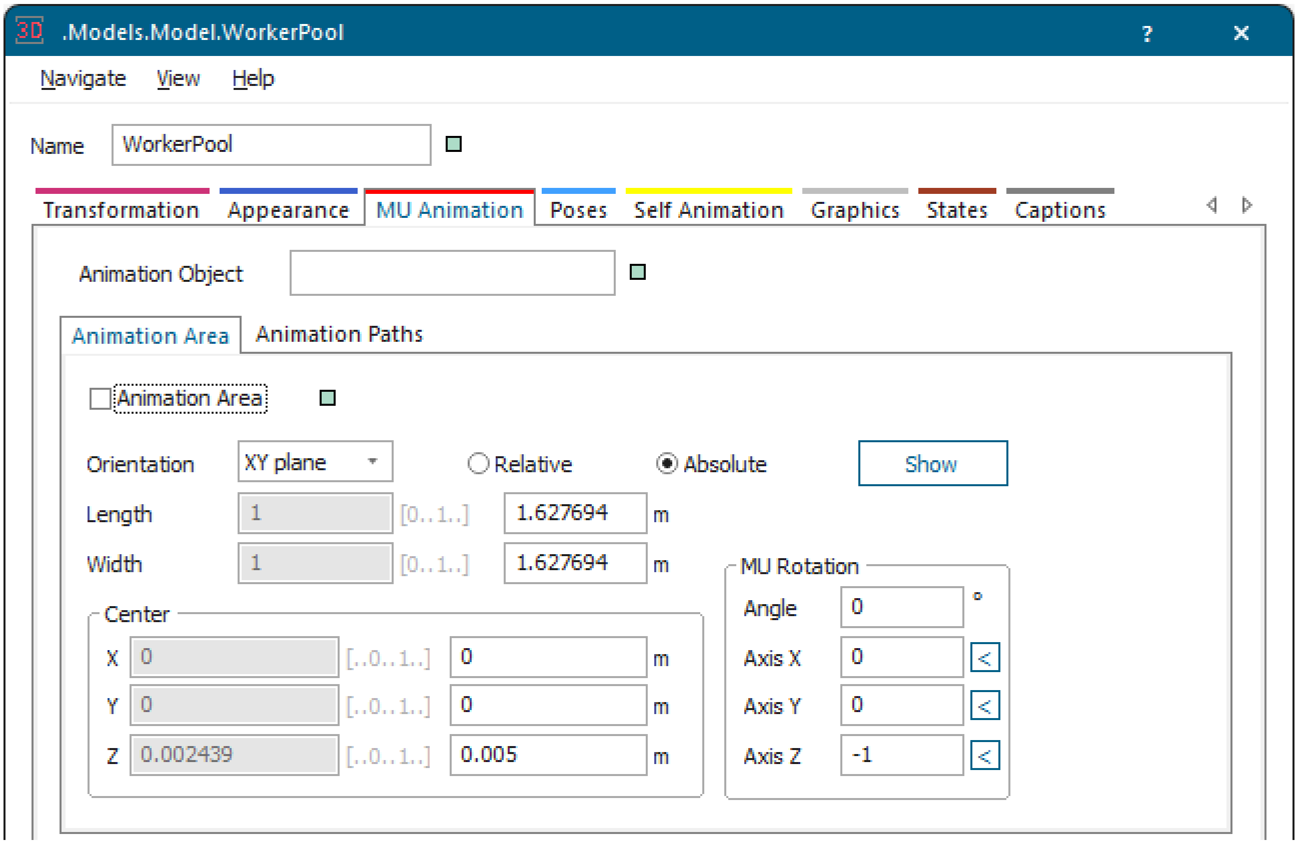
Animation paths consist of a set of points in space describing the path on which an animated moving object moves on the material flow objects or through the scene.

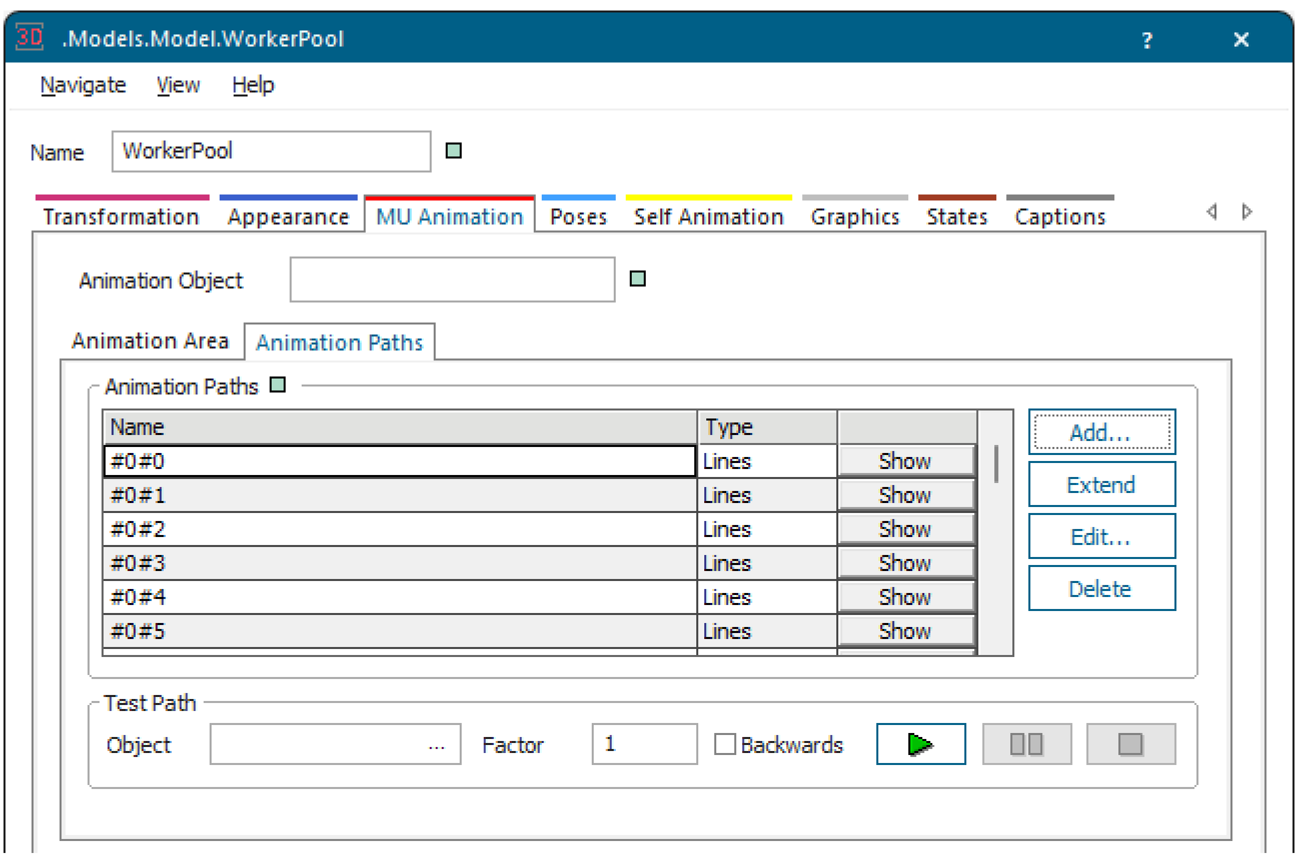
The cameras, through which you view the scene when you fly through it, also use animation paths.

The animatable objects have the animation paths described below. The procedure for creating and editing the different path types are only slightly different. For this reason we describe the common properties here.

- **MU animation paths**

  You can create and edit animation paths and the animation area, on which the parts are animated under **Edit 3D Properties** > Tab MU Animation on the **Home** ribbon tab.


.Models.Model.WorkerPool
Navigate
View
Help
Name
WorkerPool
Transformation
Appearance
MU Animation
Poses
Self Animation
Graphics
States
Captions
Animation Object
Animation Area
Animation Paths
Animation Area
Orientation
XY plane
Relative
Absolute
Show
Length
1
[0..1..]
1.627694
m
Width
1
[0..1..]
1.627694
m
MU Rotation
Angle
0
Axis X
0
Axis Y
0
Axis Z
-1
Center
X
0
[..0..1..]
0
m
Y
0
[..0..1..]
0
m
Z
0.002439
[..0..1..]
0.005
m


.Models.Model.WorkerPool
Navigate
View
Help
Name
WorkerPool
Transformation
Appearance
MU Animation
Poses
Self Animation
Graphics
States
Captions
Animation Object
Animation Area
Animation Paths
Animation Paths
Name
Type
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Lines
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#0#1
Lines
Show
#0#2
Lines
Show
#0#3
Lines
Show
#0#4
Lines
Show
#0#5
Lines
Show
Add...
Extend
Edit...
Delete
Test Path
Object
...
Factor
1
Backwards

- **Self animation paths**

  You can create and edit paths on which the object itself is animated by clicking **Edit 3D Properties** > Tab Self Animation on the **Home** ribbon tab.

  The anchor points of the self animation paths are the vertices of the **line**, **polycurve**, or **spline** on which the selected object itself moves. You can, for example, use it a self animation path to simulate the movement of a robot.

  The **polycurve animation** anchor points are the vertices of a curved path, i.e., a sequence of **curved and straight lines**, on which the material flow object itself moves.

  The **spline animation** anchor points are the vertices of a spline curve.

- **Camera animation paths**

  You can create and edit paths on which the camera moves through the 3D scene in the *Frame* by clicking **Edit 3D Properties** > Tab Camera Animation on the **Home** ribbon tab.

  The anchor points of the camera animation paths are the vertices of the **line**, **polycurve**, or **spline** on which the camera moves through the scene in the *Frame*.

  The **polycurve animation** anchor points are the vertices of a curved path, i.e., a sequence of **curved and straight lines**, on which the camera moves.

  The **spline animation** anchor points are the vertices of a spline curve.

Compare Model Joints and Poses

Back to Visualizing the Material Flow

Videos on YouTube

https://youtu.be/YleFq6afRs4?si=QsQRL-NA5J-OMGpK&t=145

https://youtu.be/jHOZhlwGhkQ?si=0AsRC64JxvckO1sM&t=623

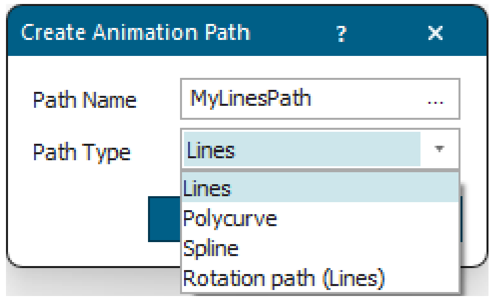
Create an Animation Path

You can create three types of animation path: A path on which the *MU* will be animated while it moves through the plant. A path on which the material flow object itself will be animated. A path on which the object camera moves through the *Frame*.

Proceed as follows:

- Open the dialog Edit 3D Properties. Click the Tab MU Animation, or the Tab Self Animation, or the Tab Camera Animation depending on the path you want to create.
- Click **Add**.
- Enter a unique **name** for the path, if you are creating an animation path. As a rule you will use the **Default** path for the animation.

- Select the **Path type** you would like to create.



  If objects, such as the *Store* or the *ParallelStation*, have several defined locations for placing incoming mobile objects, you can distribute them on the **animation area** on the tab **MU Animation**.

- Click **OK** to create the path you just defined. Initially this path consists of a single anchor point located at the origin of the selected object.
- **Extend** and/or **edit** the new initial path by clicking **Edit 3D Properties > MU Animation/Self Animation/Camera Animation > Edit > Path Anchor Points** on the **Home** ribbon tab.

Compare Insert Curved and Straight Segments

Compare Draw Straight and Curved Segments with a 90° Angle

Compare Draw Straight and Curved Segments without Fixed Values

Back to Edit an Animation Path

Videos on YouTube

https://youtu.be/YleFq6afRs4?si=QsQRL-NA5J-OMGpK&t=145

https://youtu.be/h2cFAH46ESQ?si=39K86uqevyTU2N3Y&t=870

Create an Animation Path that Rotates Objects

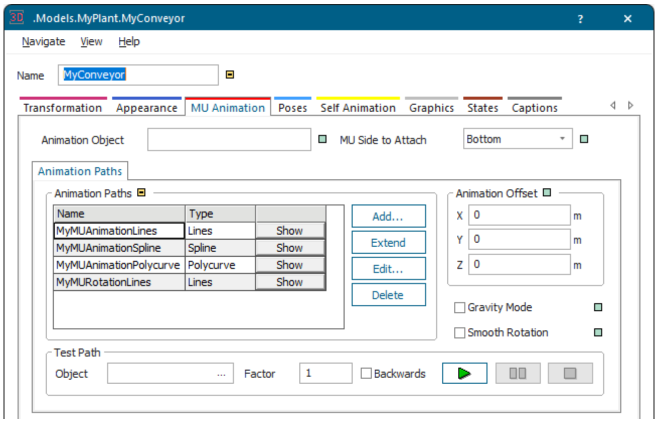
You can create an animation path, which causes a circular rotation of an object, for example the movement of the arm of a crane, by setting parameters for the rotation instead of by editing anchor points.

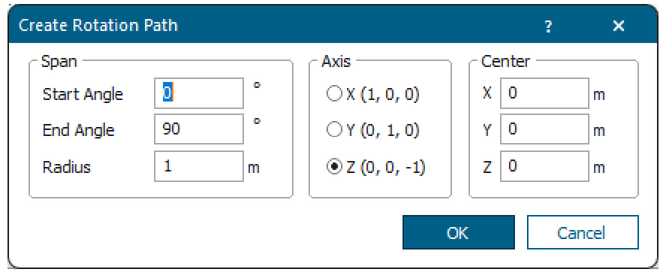
Proceed as follows:

- Select the material flow object or the scene object and click **Edit 3D Properties** on the **Home** ribbon tab.
- To rotate the *MUs* that the selected object processes, click the **Tab MU Animation**.

  To rotate the *object itself*, click the **Tab Self Animation**.

- Click **Add**. Enter a name for the rotation path into the dialog **Create Rotation Path** that opens. Select the **Path Type** > **Rotation Path [MU animation] - Lines**.





- Edit the settings in the dialog **Create Rotation Path**:

  Enter the **Start Angle** where the rotation starts. Enter the **End Angle** where the rotation ends. *3D* only rotates the *MU* if these values are not identical. The values can be greater than 360 degrees and smaller than -360 degrees to execute more than an entire rotation.

  Start Angle and **End Angle** set the direction of the rotation, depending on which value is the greater one:

  - An **End Angle** greater than the **Start Angle** results in a clockwise rotation.
  - An **End Angle** smaller than the **Start Angle** results in a counter-clockwise rotation.
- Enter the values for the **Axis** around which you want to rotate the *MUs*.
  - Normally, you will use the **Z-Axis** (0,0,1) to rotate the object on the **XY** plane.
  - You also can define the **Y-Axis** (0,1,0) to rotate the object on the **XZ** plane or the
  - **X-Axis** (1,0,0) to rotate the object on the **YZ** plane.

    Naturally you can also define any other axis to rotate the object diagonally in 3D space.

    To tilt the rotation axis, and thus the way the *MU* rotates, you would enter a value into at least two text boxes.

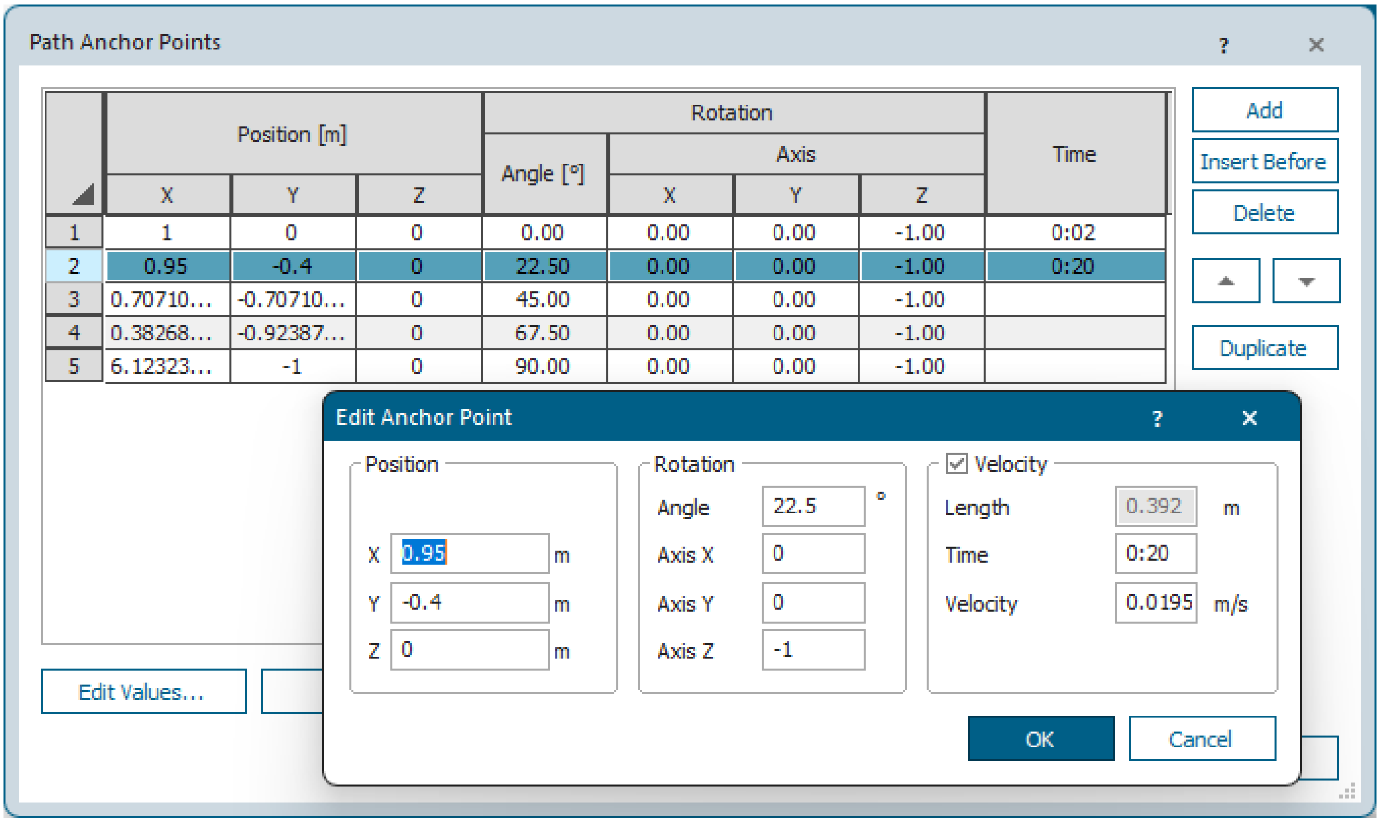
    You can change one, two or three settings to tilt the axis the *MU* rotates around on the surface of the object.
- Enter the **Center** point of the rotation in the coordinate system of the selected object. If the center is not the same as the start position, the rotated object moves on a circular arc and stops at a different position than its starting point, except after a 360 degree rotation.
- Click **Apply** to compute the animation path. This path is the trace on which the animated object moves. The movement on this path is the rotation.

  > **Note**
  >
  > You can use the method _3D.SelfAnimations.scheduleRotation to run **self-rotations** or **camera-rotations** without having to create a path.

- To test the rotation path you defined, select an **object** and start the test animation.

- If the rotation movement does not meet your expectations, open the dialog **Create Rotation Path** again, and edit the parameters there. Then, click **Apply** and test the settings again until the rotation meets your needs.
- To change the duration of one or more individual rotation steps:
  - Select the path in the list, and click **Edit**.
  - Select the anchor point in the dialog **Path Anchor Points** and click **Edit Values** to open the dialog **Edit Anchor Point**.

  Under **Time** you can edit the **time** of every anchor point, which matches the rotation steps.



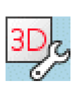
Compare Rotation path [camera animation] - Lines

Back to Edit an Animation Path

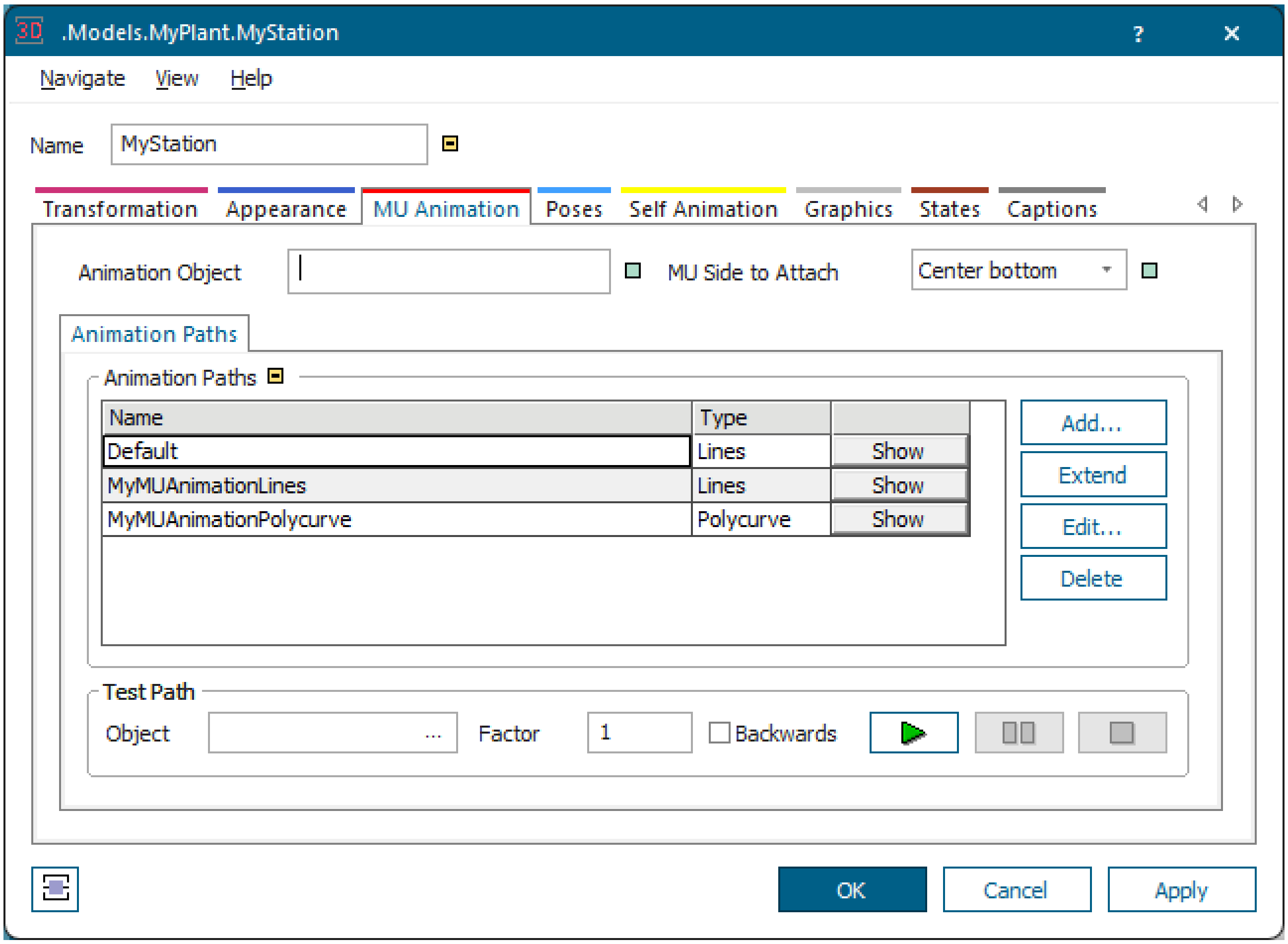
Edit an Animation Path

You can edit the different animation paths in the dialog **Edit 3D Properties**.

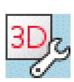
To work with **animation paths of parts (MUs) on an object**, do one of the following:

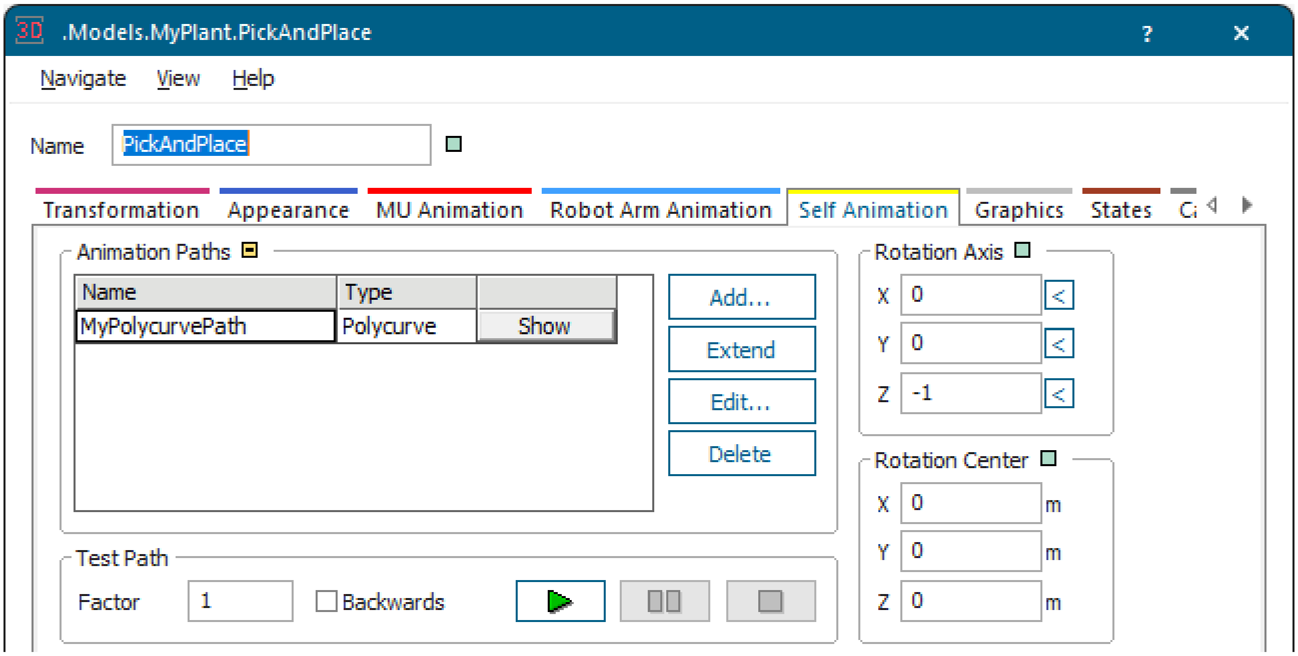
- Select the object and click  **Edit 3D Properties** on the **Home** ribbon tab.

- Click the Tab MU Animation in the dialog **Edit 3D Properties** and create a new animation path or edit an existing animation path.

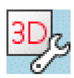


To work with **animation paths on which the object itself is animated in the scene**, do one of the following:

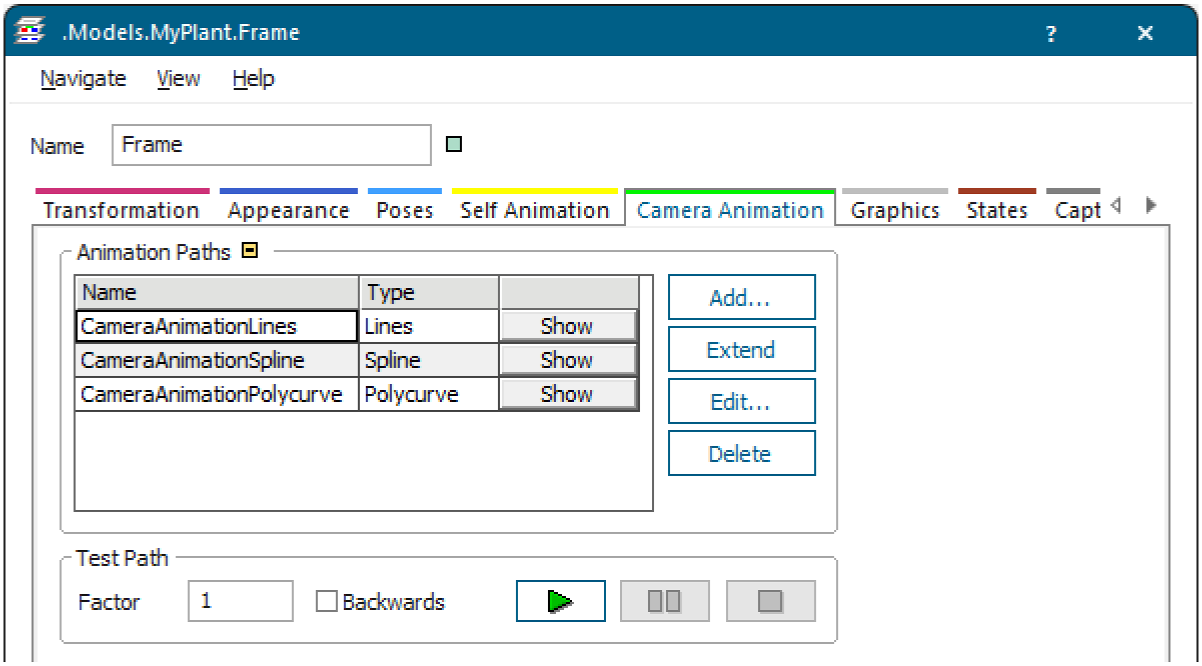
- Select the object and click  **Edit 3D Properties** on the **Home** ribbon tab.
- Click the Tab Self Animation in the dialog **Edit 3D Properties** and create a new animation path or edit an existing animation path.



To work with **camera animation paths of the Frame**, do one of the following:

- Select the *Frame*, and click  **Edit 3D Properties** on the **Home** ribbon tab.
- Click the Tab Camera Animation in the dialog **Edit 3D Properties** and create a new camera animation path or edit an existing camera animation path.

> **Note**
>
> Only the *Frame* provides the tab **Camera Animation**.



In or from the dialog you can:

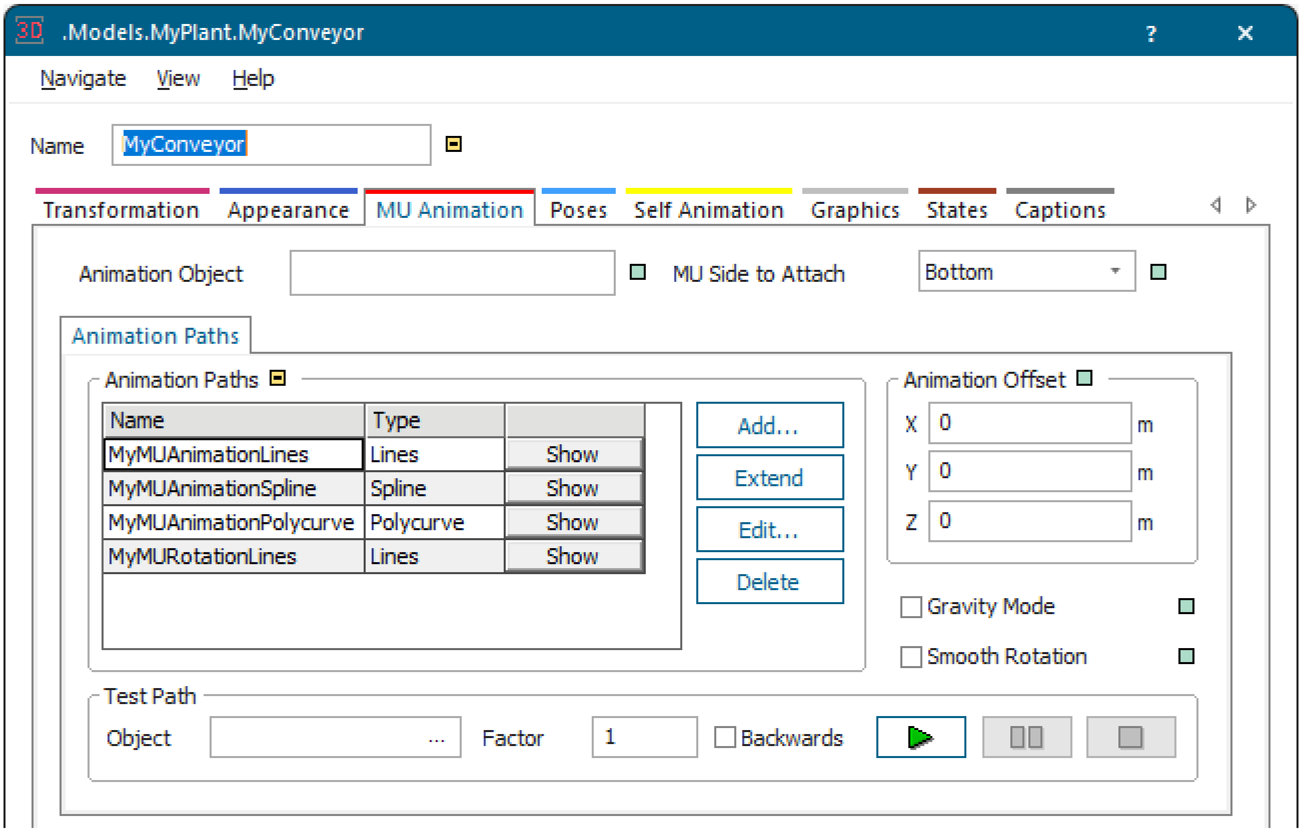
- **Create an Animation Path**.
- **Test an Animation Path**.
- **Edit an Animation Path with the Mouse**.
- **Edit an Animation Path via Anchor Points**.
- **Create an Animation Path that Rotates Objects**.
- **Rename** an animation path.
- **Delete** an animation path.

Back to Work with Animation Paths

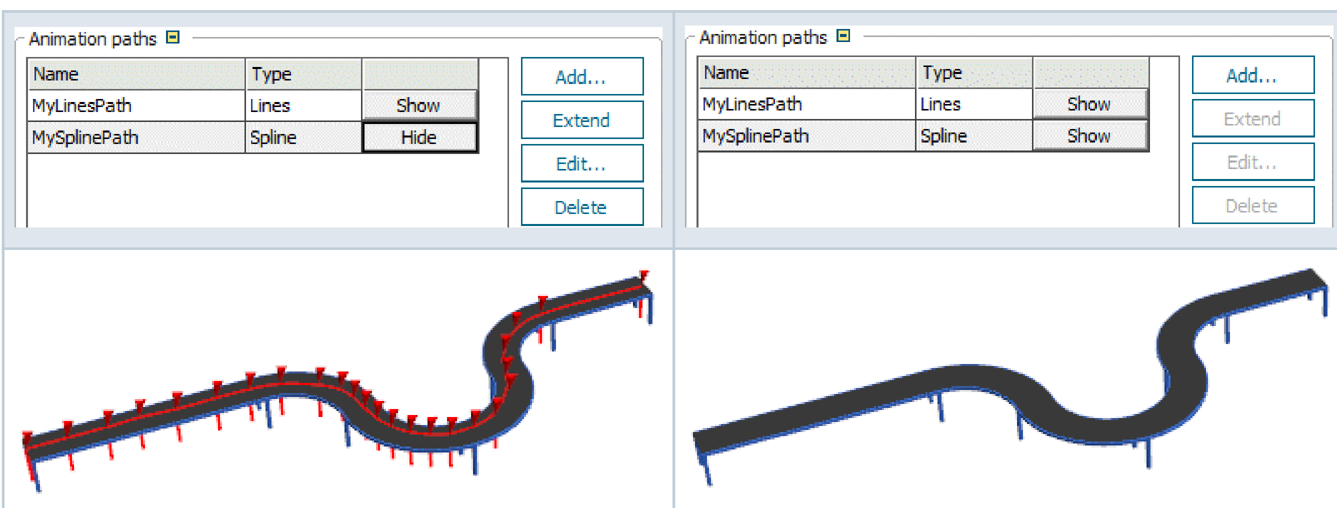
Edit an Animation Path with the Mouse

You can edit an animation path with the mouse in the *Frame* window.

- To edit an animation path, open the dialog **Edit 3D Properties**. Click the tab **MU Animation**, the tab **Self Animation**, or the tab **Camera Animation [Frame]**, or the tab **Robot Arm Animation [PickAndPlace]** depending on the path you want to edit.



- To show the selected path in the scene window, click the button in the cell **Show** under **animation paths**. *Plant Simulation* then shows the path with its anchor points.



- To add new anchor points to the path, click **Extend** and click the left mouse button at the position where you would like to insert a new anchor point. Repeat this as often as you need, until the path has the shape you want.
- To change the shape of the path, select an anchor point and manipulate it with the mouse and/or use the settings by clicking **Edit 3D Properties** > Tab Transformation on the **Home** ribbon tab.

  When you drag an anchor point in the X-direction/Y-direction, *Plant Simulation* also drags the connected curved segments in both directions (until the next straight segment follows).

- To rotate an anchor point, click it, hold down the left mouse button, and drag the mouse, and/or click **Edit 3D Properties > Transformation**.

> **Note**
>
> When you rotate a anchor point describing an animation path, this also rotates any *MU* moving along that path.

- To delete a part of the path, select an anchor point and press **Delete**. This removes the selected anchor point from the path, thus modifying its shape.

Back to Work with Animation Paths

Edit a Curved Path with the Mouse

You an edit and extend a curved path with the mouse in a number of ways.

In addition to editing any of the path types with the mouse, you can edit polycurve paths as follows:

- **Add new anchor points** to the animation path by clicking **Extend**.
  - To add a **straight segment**, click the left mouse button at the position where you would like to insert a new anchor point.
  - To add a **curved segment**, hold down **Ctrl**, and click the left mouse button at the position where you would like to insert a new anchor point.
  - Repeat this as often as you need, until the path has the shape you want.
- Change the shape of a straight segment by selecting an anchor point and by dragging the mouse.

  When you drag an anchor point in the X/Y-direction, *3D* also drags attached curved segments (until the next straight segment follows) in both directions.

Compare Model Differences in Height Between Conveying Elements

Compare Insert Curved and Straight Segments

Compare Draw Straight and Curved Segments with a 90° Angle

Compare Draw Straight and Curved Segments without Fixed Values

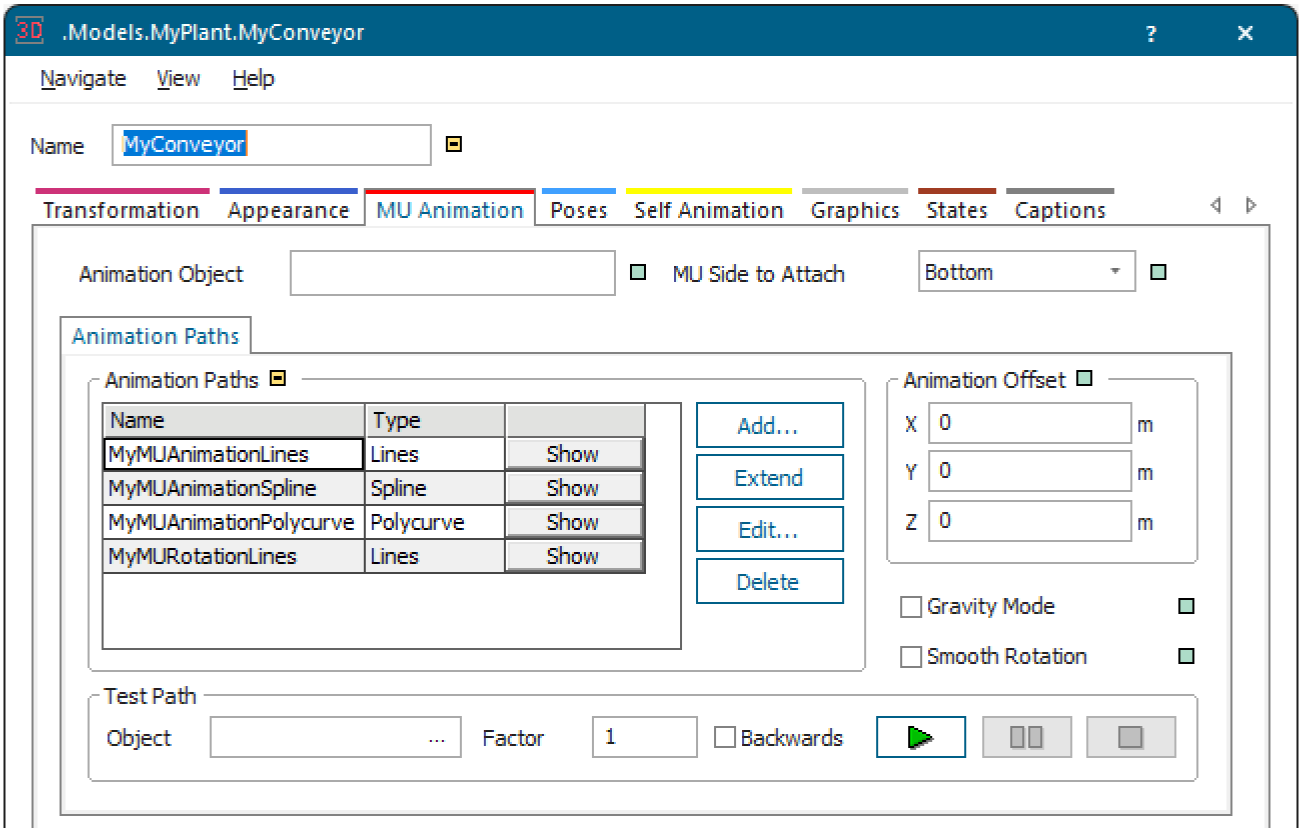
Back to Work with Animation Paths

Edit an Animation Path via Anchor Points

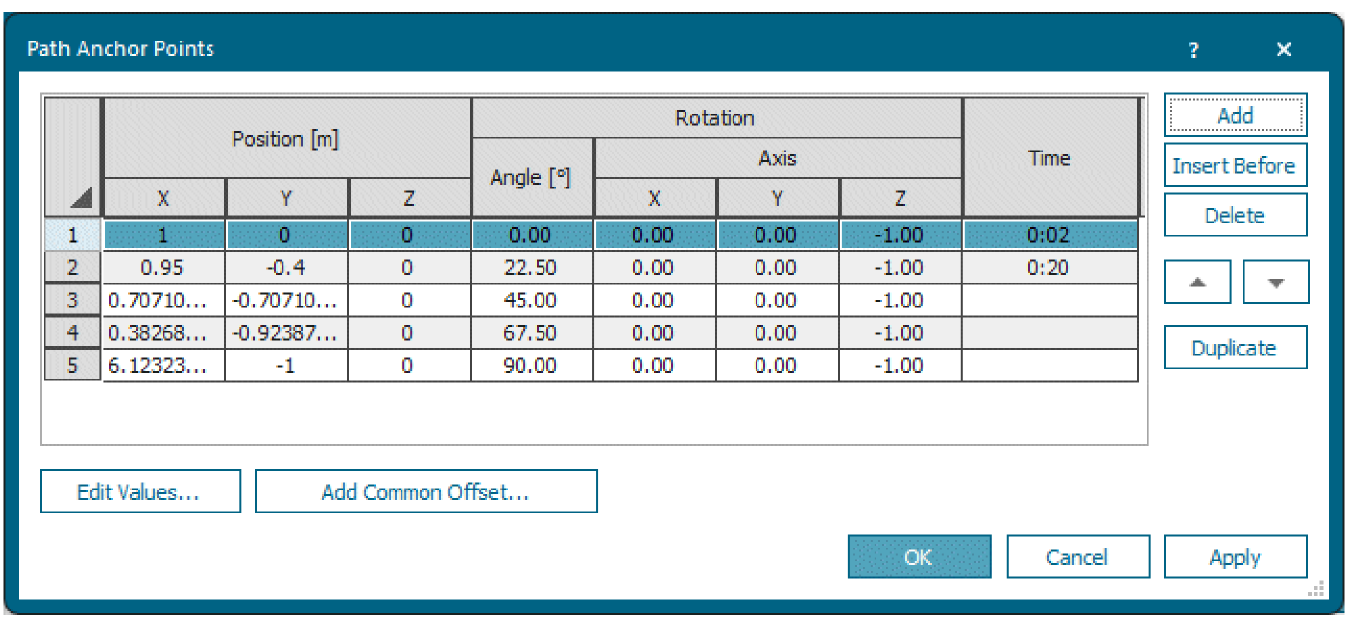
You can edit an animation path by entering exact values into the dialog **Path Anchor Points**.

At times it may not be precise enough to **Edit an Animation Path with the Mouse**. Then, you can enter exact values into the dialog **Path Anchor Points**:

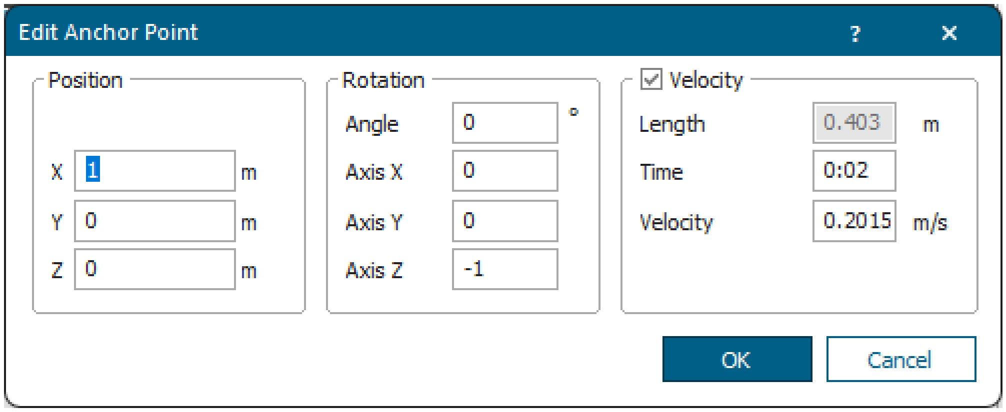
- Open the dialog **Edit 3D Properties**. Select the animation path you would like to edit on the **Tab MU Animation**, on the **Tab Self Animation**, or on the **Tab Camera Animation**.
- Click **Edit**.



- Select the anchor point you would like to edit and in the dialog **Path Anchor Points** and then click the button **Edit Values** or double-click the row in the table.



- Enter or select the new settings for the **Position** and for the **Rotation** of this anchor point into the dialog **Edit Anchor Point**.



- To add a new anchor point to the end of the selected path, click **Add**. The list adds the settings of the new anchor point you added to its bottom. Once you have added an anchor point, you can edit its settings in the dialog **Edit Anchor Point**.
- To insert a new anchor point before the selected anchor point, click **Insert Before**.
- To delete an anchor point, select it in the list, and click **Delete**.
- To delete contiguous anchor points, hold down **Shift** and click the first anchor point in the range and then the last anchor point.

  To delete non-contiguous anchor points, hold down **Ctrl** and click the anchor points you want to delete one after the other.
- To move the selected anchor point up one position in the list, click **Move Up**.
- To move the selected anchor point down one position in the list, click **Move Down**.
- To duplicate the settings of the selected anchor point and to insert the duplicated anchor point below the original, click **Duplicate**.

> **Note**
>
> You can edit paths of type **Polycurve** in the dialog **Edit**.

Back to Edit an Animation Path

Test an Animation Path

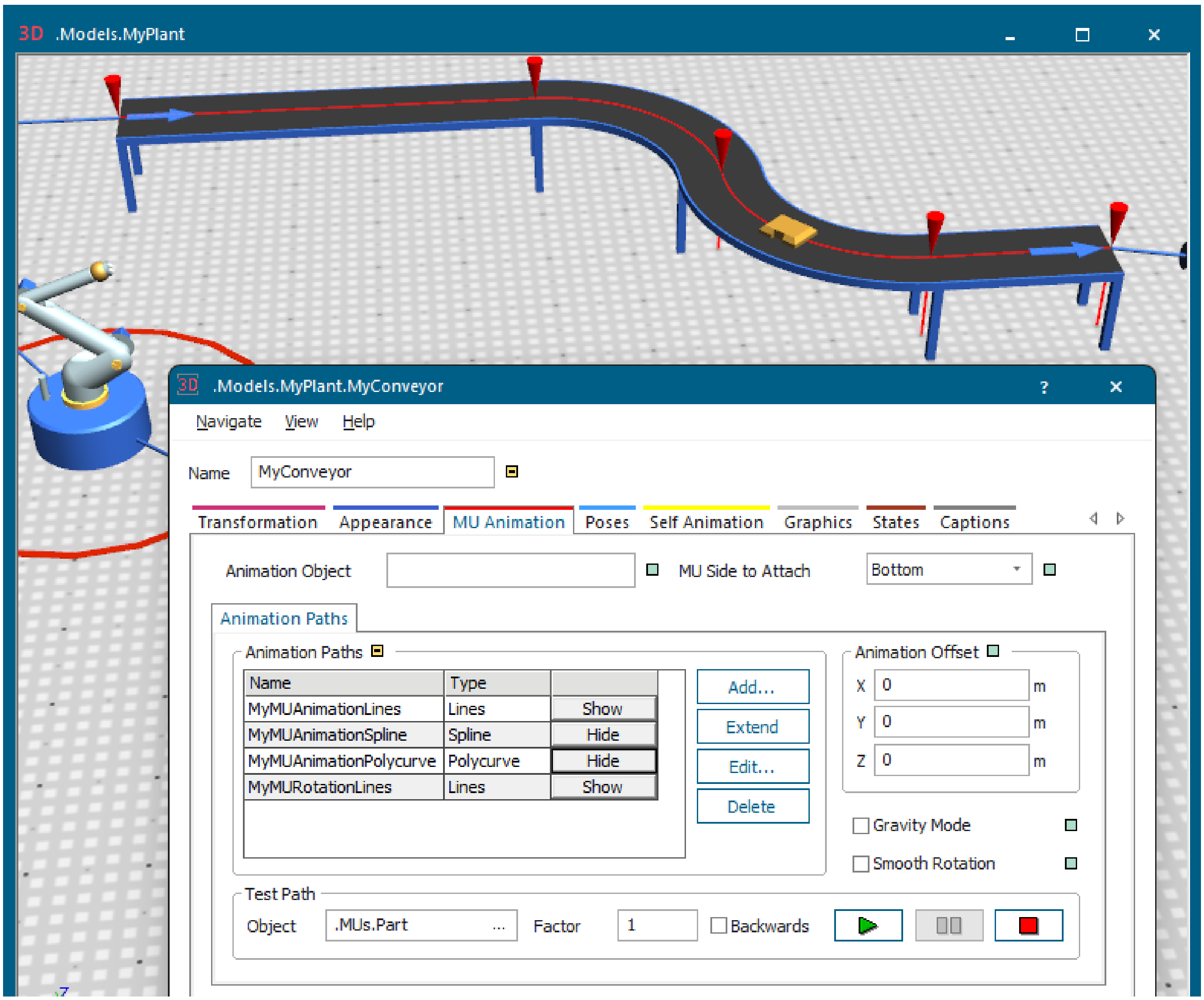
After defining the animation path, you can test the settings you chose on the **Animation** tabs in the group box **Test Path**, and modify them, if the results do not meet your expectations. This way, you do not have to run a simulation to test your settings.

Proceed as follows:

- Select the **Test Object**, i.e., a *MU* whose graphic *3D* copies and uses as the test object.

- Enter the **Velocity** with which the test object moves on the path.
- Select if the test object moves in the forward direction or if it moves **Backwards**.

  3D automatically enters a negative **Velocity**, when you select **Backwards**.
- Click to play the animation with the settings you selected.
- Click to pause the test animation.
- Click to stop the test animation.



Back to Work with Animation Paths

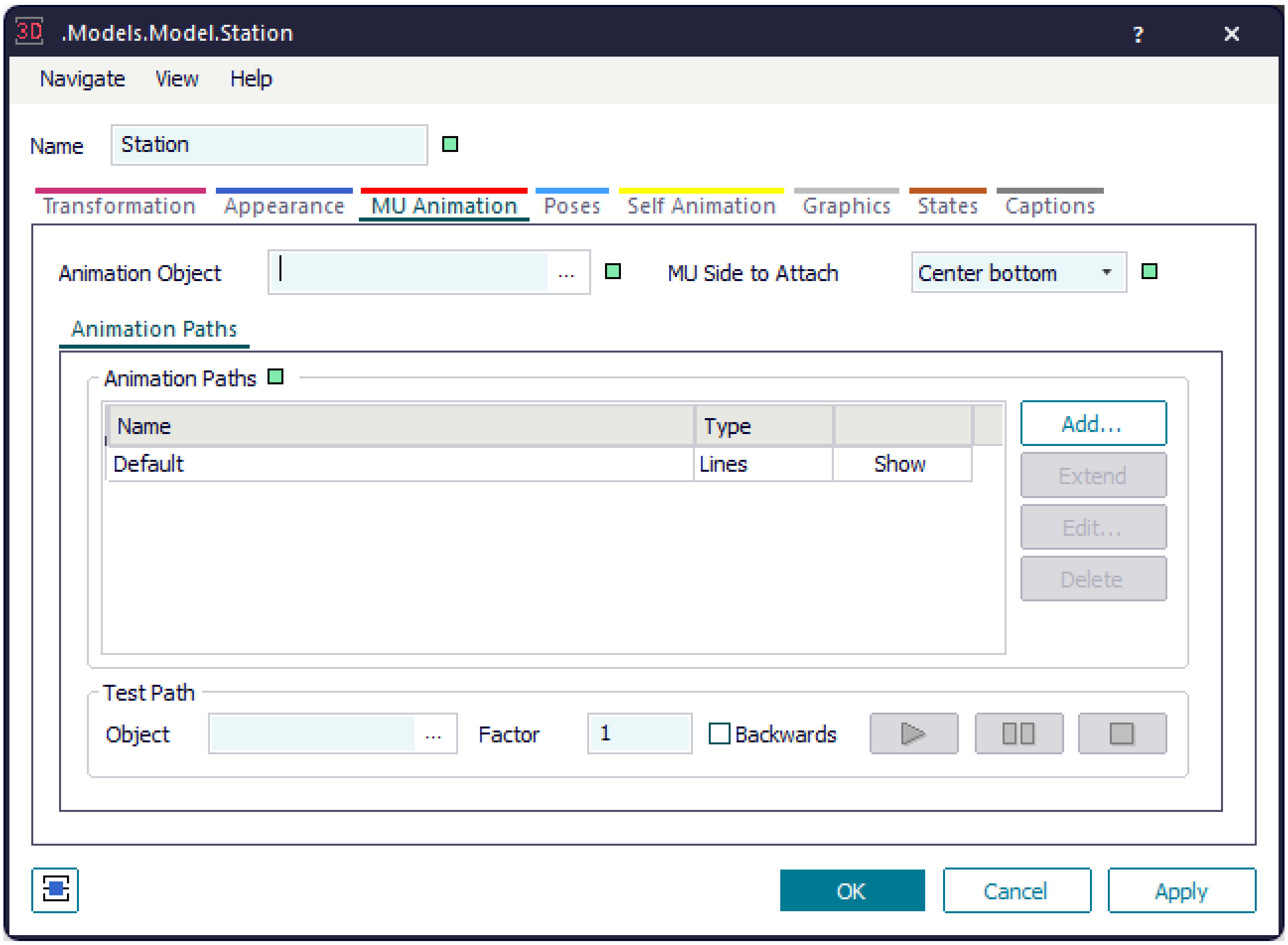
Controlling the MU Animation

Control the MU Animation

On the **tab MU Animation** you can define how the mobile objects (*MUs*) are animated on the material flow objects. The *Frame* also counts as a material flow object.

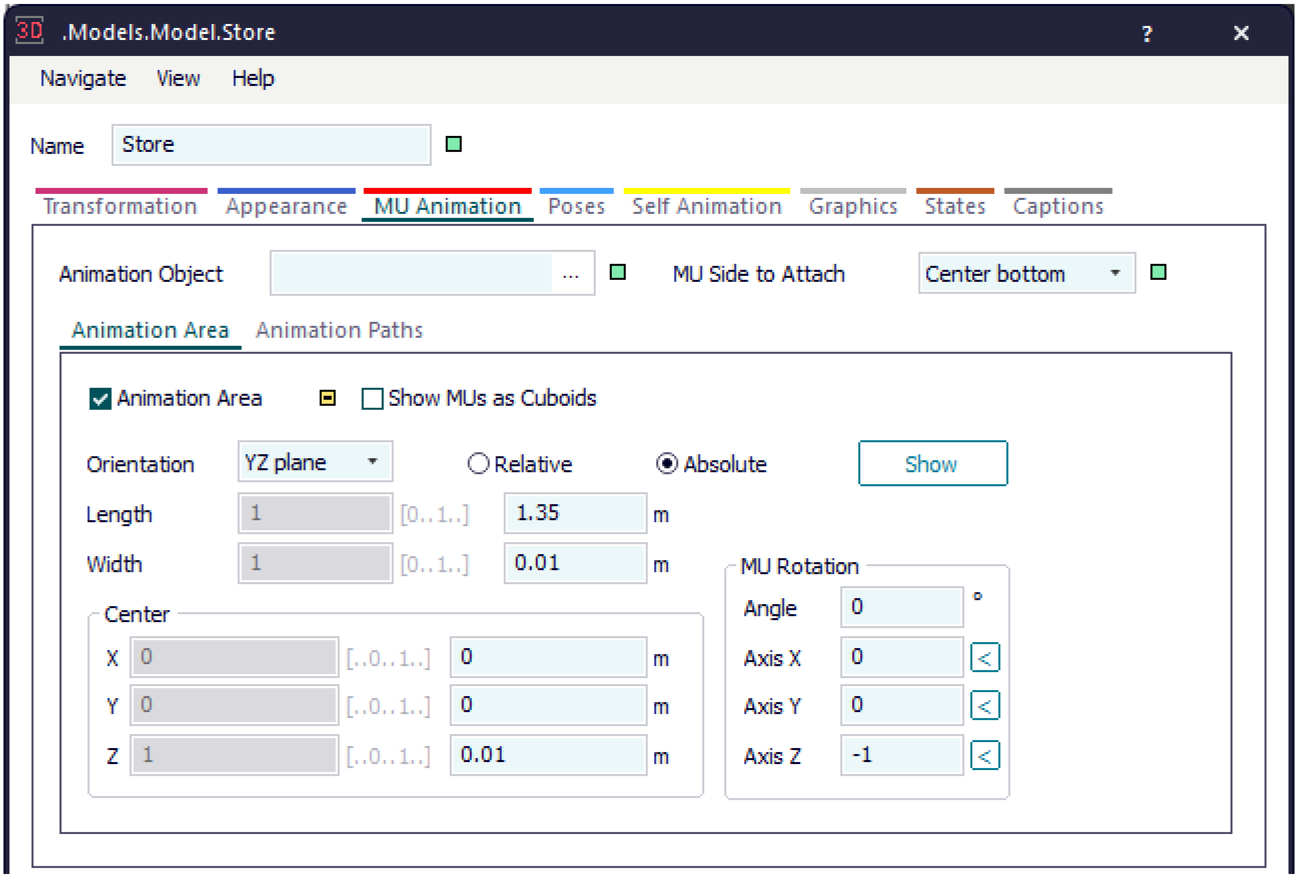
The tab looks different for *objects without loading space* and for *objects with loading space*.

- On the **Sub-tab Animation Paths** you can define these **MU Animations**:



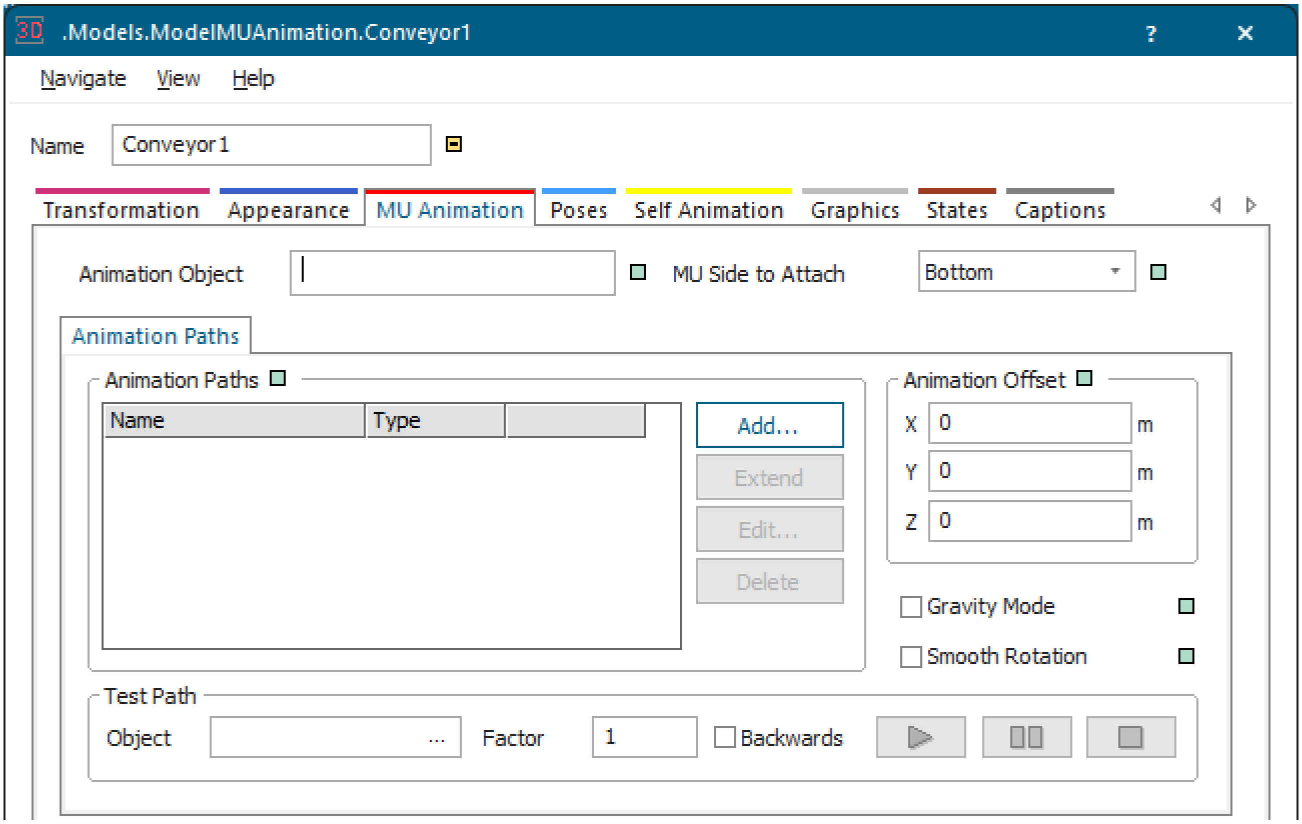
  - The **MU Animation on Length-Oriented Objects**
  - The **MU Animation on Point-Oriented Objects**
  - The **MU Animation on Point-Oriented Objects Holding Several Parts**
  - The **MU Animation on Animation Paths on the Workplace**
- On the **Sub-tab Animation Area** you can define the **MU Animation on Objects with Loading Space** by activating the **Animation Area**.

The **objects with a loading space** – **AGVPool**, **Container**, **ParallelStation**, **PlaceBuffer**, **Store**, **Transporter**, **Worker**, **Workplace**, and **WorkerPool** – use the **Animation Area**, provided you activated it. If it is not activated, the objects use **animation paths**. Except for the *AGVPool* you have to create the animation paths yourself though.



MU Animation on Length-Oriented Objects

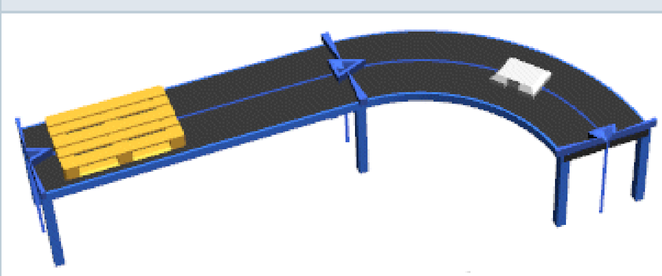
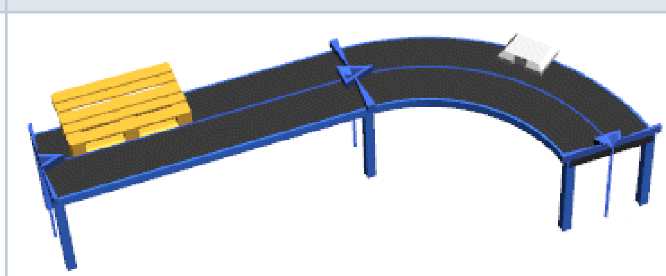
The **length-oriented objects** animate the parts automatically. If a path named **Default** exists, the objects use that path.



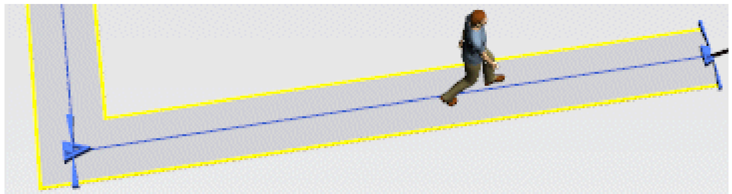
- The **length-oriented objects** *AngularConverter*, *FootPath*, *Turnplate*, and *Turntable* can each hold and transport a single part.
- The **length-oriented objects** *Conveyor*, *Track*, and *TwoLaneTrack* can hold and transport several parts.

For all **length-oriented objects** you can add an **Animation Offset** and select or clear **Gravity Mode**. To do so, press the **M** key to show the manipulators in the scene.

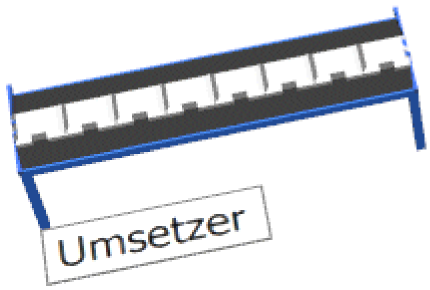
- The **Conveyor** animates the parts on the blue line in the center of the conveyor. The standard setting looks like this:

Animation offset up is 0	Animation offset up is 0.4
	

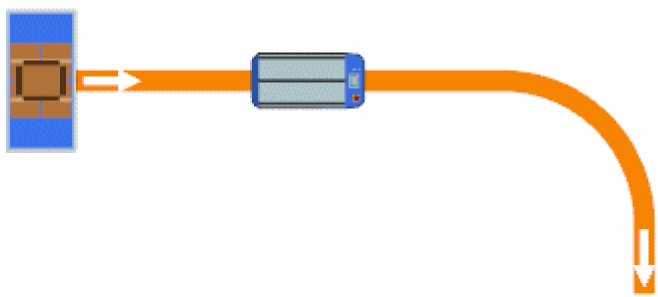
- The **FootPath** animates the *Workers* on the blue line in the center.



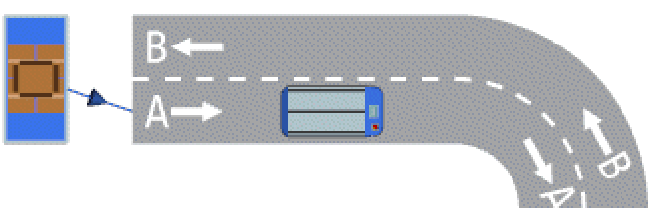
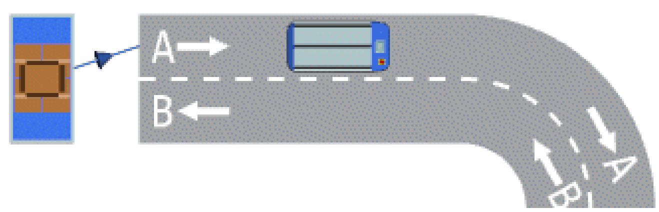
- The **Converter** animates the parts automatically. If a path named **Default** exists for length-wise movements and **Cross** for cross-wise movements exits, it uses that path.



- The **Track** animates the *Transporters* in the center of the travel track.



- The **TwoLaneTrack** animates the *Transporters* automatically. If paths named **A** or **B** exist, depending on the lane and the setting for **right-hand traffic** or **left-hand traffic** exist on which the *Transporter* drives, it uses that.

Right-hand traffic	Left-hand traffic
	

Back to **Control the MU Animation**

Video on YouTube

https://youtu.be/rQi5oRyZckQ?si=nQLvgpv7FieRe5mX&t=455

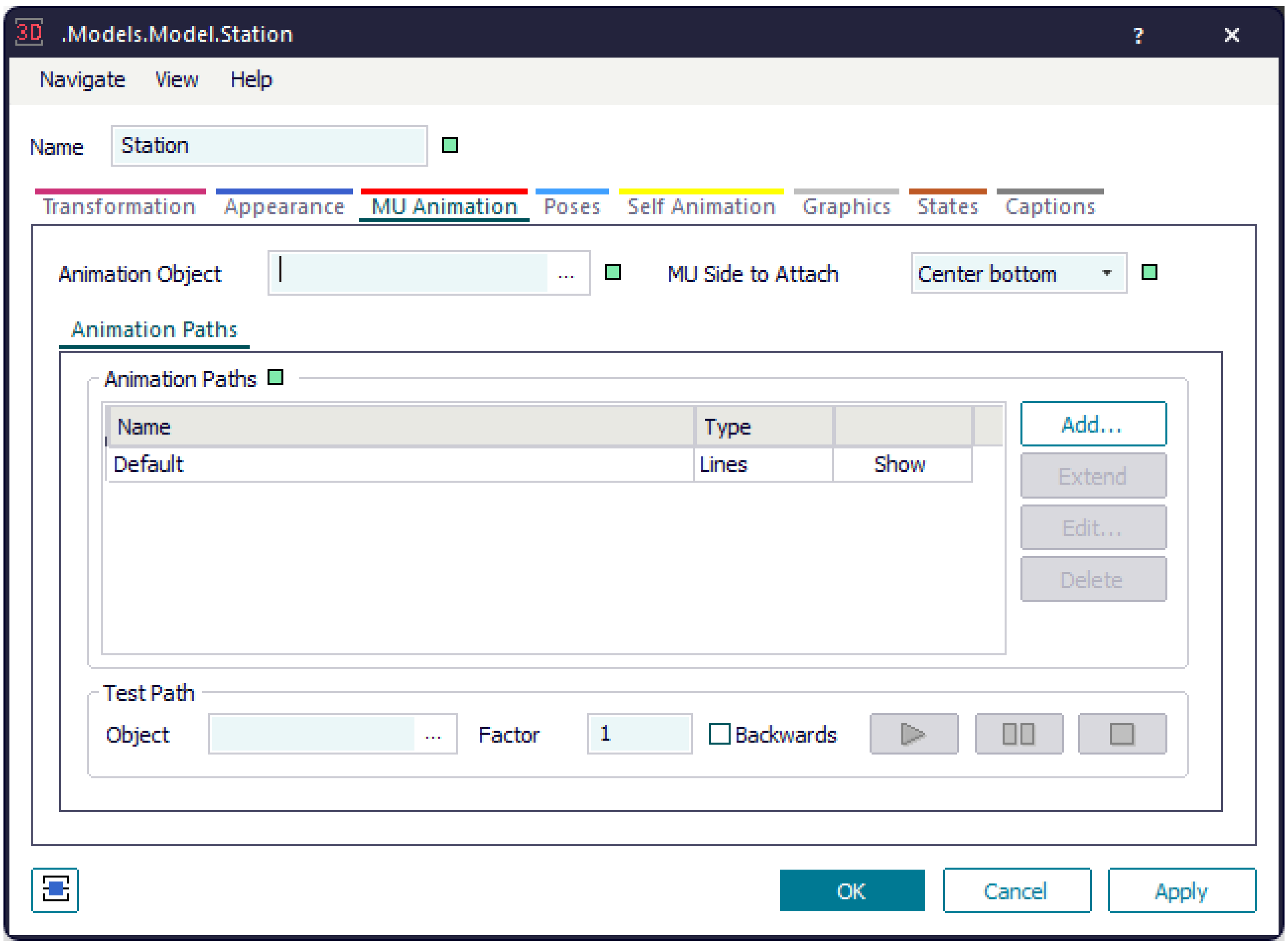
MU Animation on Point-Oriented Objects

You can create an MU animation path on point-oriented objects.

For **point-oriented objects, which can hold a single part**, for example the *Station*, you can:

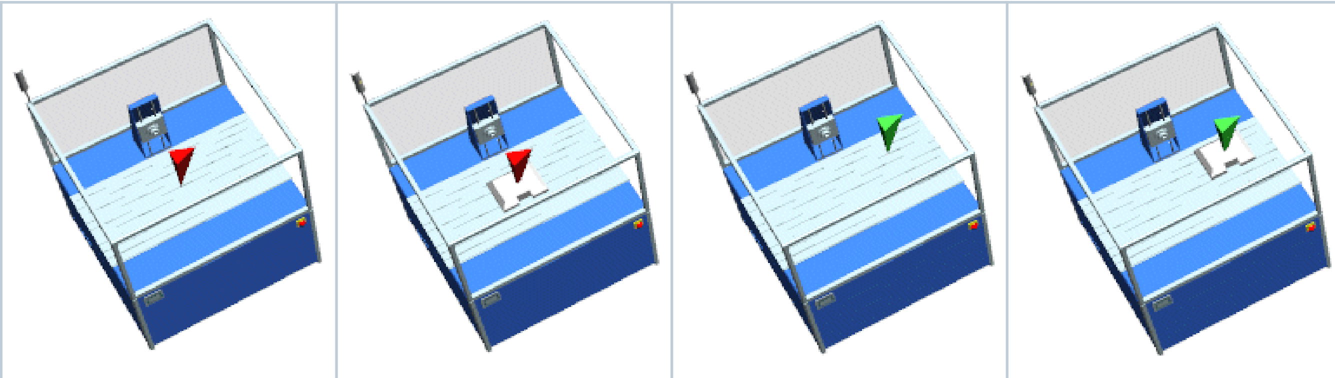
- Specify the **Animation Object** and the **MU Side to Attach**.
- **Add**, **extend**, **edit**, and **delete** an animation path.

The **point-oriented objects, which hold a single part** animate the parts as follows:

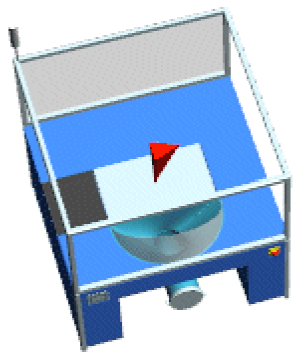


- The **Station** uses the path named **Default** for animating the parts.

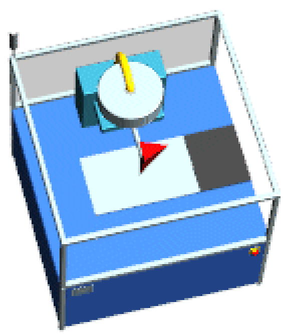
  Click **Show** to show the **MU animation path** of the *Station* in the model. As the *Station* can only process a single part, the animation path consists of a single point. The path tool, i. e., the red pyramid, shows where the part is going to be animated. You can click the path tool and move it with the arrow keys on the keyboard. The part will then be animated at the new position.



- The **DePortioner** uses the path named **Default** for animating the parts.



- The **Portioner** uses the path named **Default** for animating the parts.



Back to Control the MU Animation

Video on YouTube

https://youtu.be/rQi5oRyZckQ?si=GwvsYiXqtDPQHw-8&t=76

MU Animation on Point-Oriented Objects Holding Several Parts

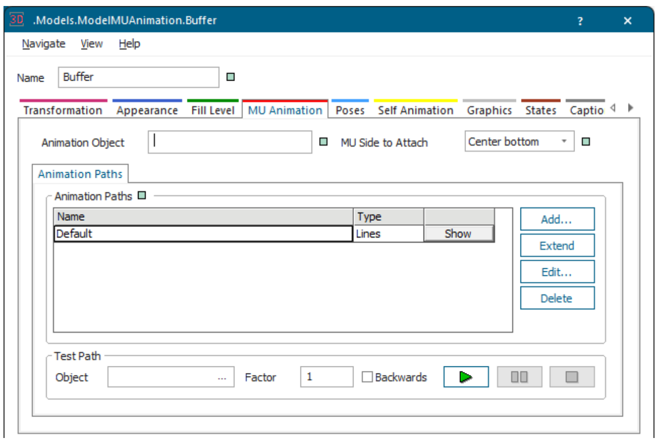
You can create an MU animation path on point-oriented objects holding several parts.

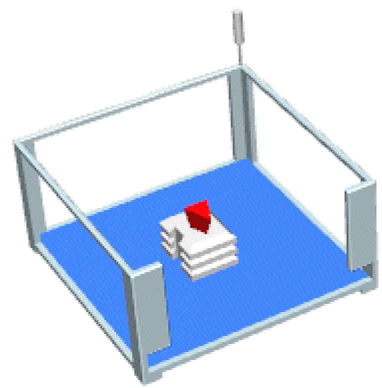
For **point-oriented objects, which can hold several parts**, you can:

- Specify the **Animation Object** and the **MU Side to Attach**.
- **Add**, **extend**, **edit**, and **delete** an animation path.

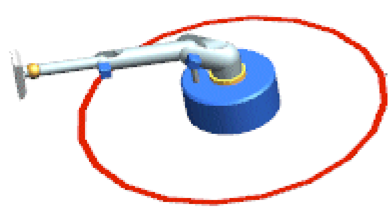
The **point-oriented objects, which can hold several parts** animate the parts as follows:

- The **Buffer** stacks the parts along the standard animation path upward one on top of the other. You can set the capacity of the *Buffer* in the dialog of the simulation properties on the tab **Attributes**.



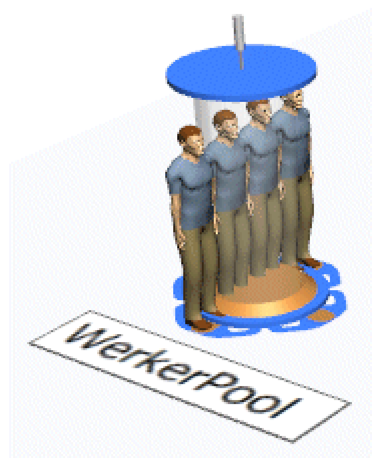


- The **PickAndPlace robot** animates the parts automatically. If a path exists, it uses the first path for the first *MU*, the second path for the second *MU*, etc. If no suitable path exists, but if a path named **Default** exists, the *robot* uses that path.

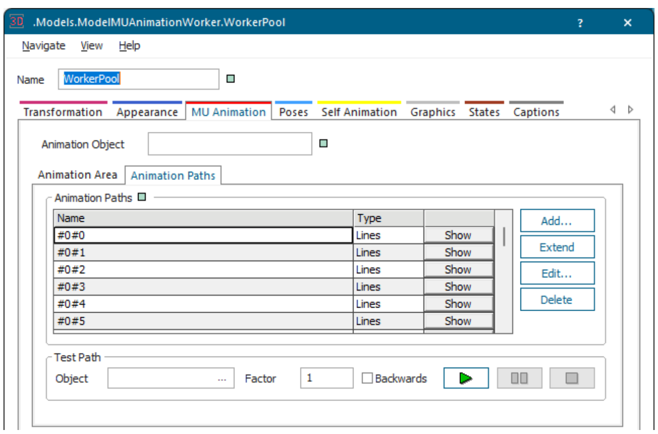


- For the **WorkerPool** you can select if it shows the *Workers* in the *WorkerPool* on **animation paths** or on an **animation area distributed along the Y-dimension**.

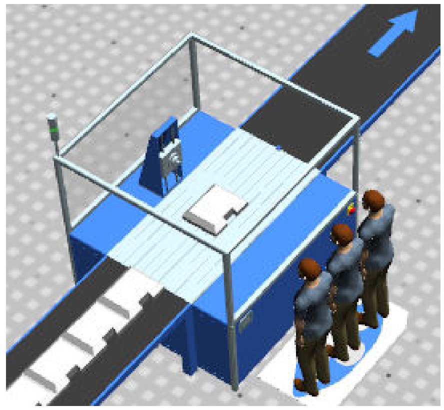
  The *WorkerPool* in the *Class Library* only shows the created *Workers* when you activate **Show Content** on the tab **Graphics**. In our example we created four *Workers*.

MU-Animation > Animation Path	MU-Animation > Animation Area
	

For the setting **Animation Path** the *WorkerPool* uses the first path for the first *MU* for the animation, the second path for the second *MU*, etc. If no suitable path exists, but if a path named **Default** exists, the *WorkerPool* uses that path.



- For the **Workplace** you can select if it shows the *Workers* on it on **animation paths** or on an **animation area distributed along the X-dimension**. Our three *Workers* on an **animation area** look like this:



If you do not use an animation area this applies:

Plant Simulation animates the first *Worker* on the first animation path, the second *Worker* on the second animation path, and the third *Worker* on the third animation path. You have to define these animation paths yourself. If, for example, no third path exists for the third *Worker*, *Plant Simulation* uses the **default** animation path. If no default animation path exists, *Plant Simulation* does not animate the *Worker* at all.

CompareHow Workers Queue Up in Front of the Workplace

Back to Control the MU Animation

MU Animation on Objects with Loading Space

Plant Simulation animates *MUs* on objects with a loading space on their **animation area**.

The **objects with a loading space** – AGVPool, Container, ParallelStation, PlaceBuffer, Store, Transporter, Worker, Workplace, and WorkerPool – use the animation area, provided you activated it. If it is not activated, the objects use animation paths.

> **Note**
>
> The indexes for paths start with 0, not with 1. The signature **#0#0** designates the first animation point, **#1#2** designates the position with the X-index 2 and the Y-index 3, etc.

As a rule you can use the factory settings for these objects unless these do not meet your requirements. *Plant Simulation* distributes the parts evenly across the animation area, which is defined by its length and its width.

> **Note**
>
> We recommend to use the **Animation Area** instead of animation paths as this means substantially less modeling effort on your side because you do not have to create the animation paths.

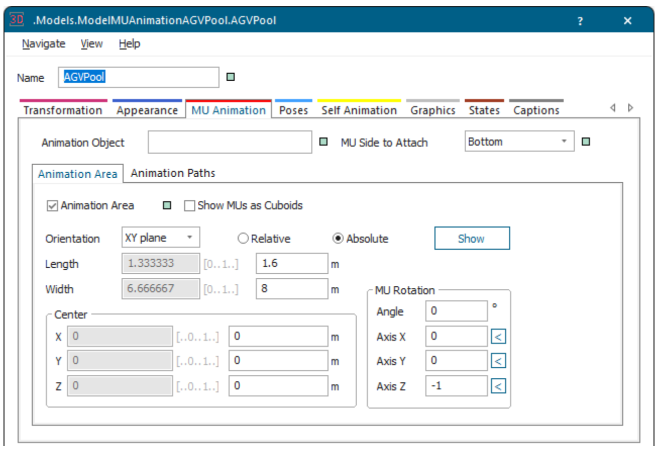
The **objects with a loading space** which use the Animation Area animate the parts as follows:

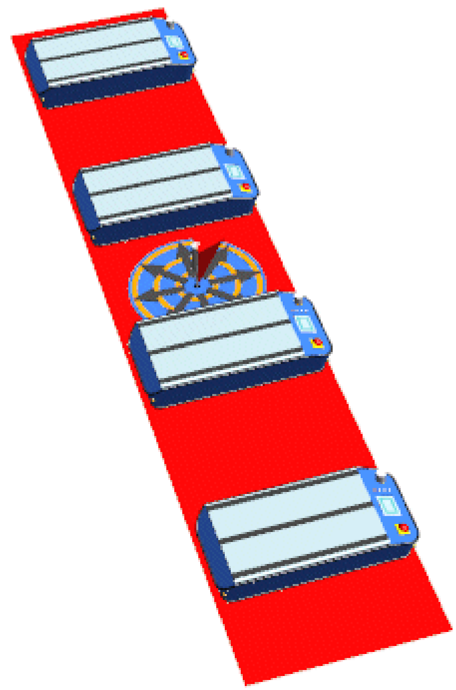
- The **AGVPool** only shows the created *Transporters* if you activate **Show Content** on the tab **Graphics**.

  The *AGVPool* animates parts as follows:

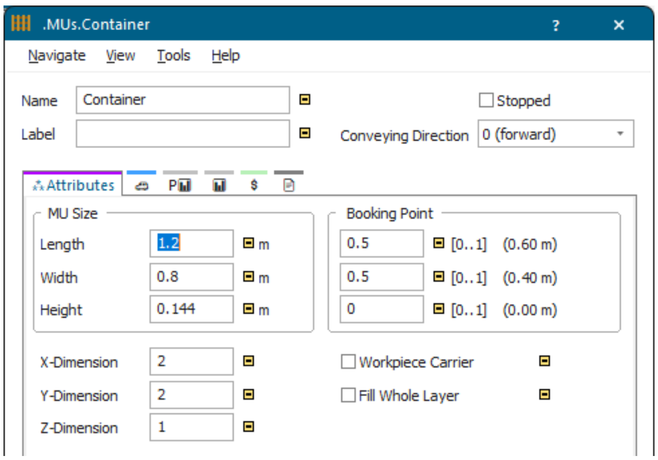
  - If the **Animation Area** is active, the *AGVPool* uses it. Here the index of the *MU* is the index of the area in the Y-direction; the capacity in the X-direction always is 1.
  - If the **Animation Area** is not active, the *AGVPool* uses the first path for the first *MU* for the animation, the second path for the second *MU*, etc. If no suitable path exists, but if a path named **Default** exists, the *AGVPool* uses that path.

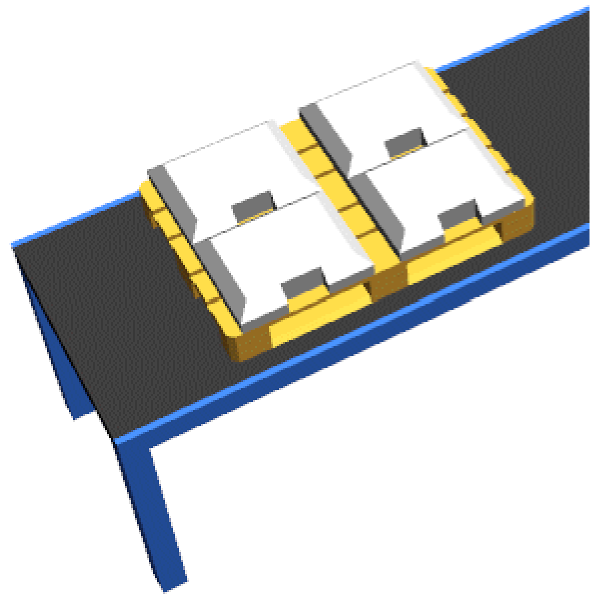
  In our example we created four *Transporters*.



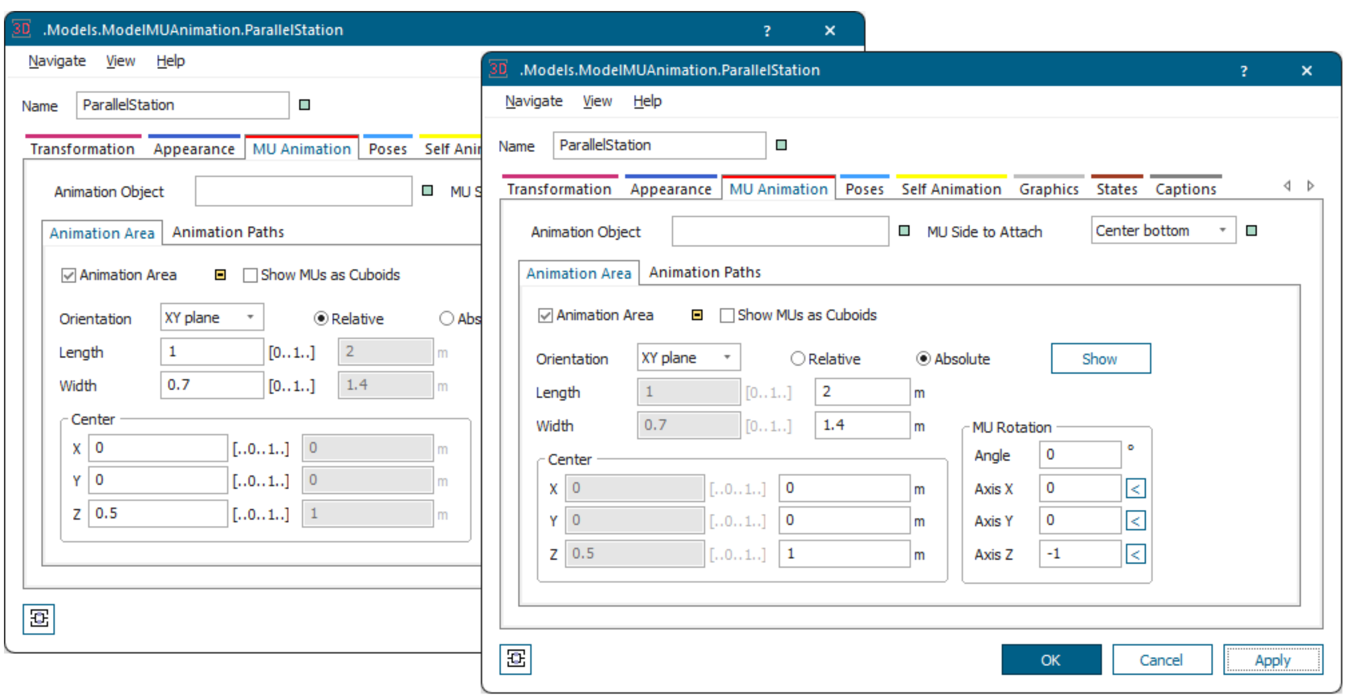


- The **Container** with the standard setting, two parts each in two rows, looks like this with four loaded parts.

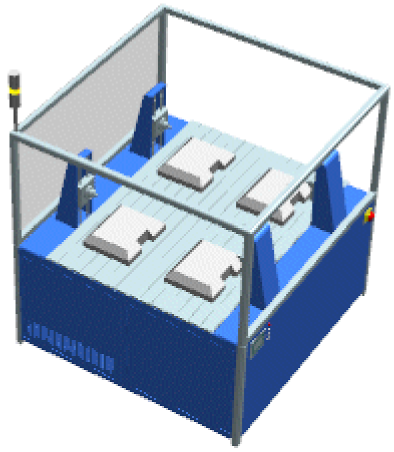
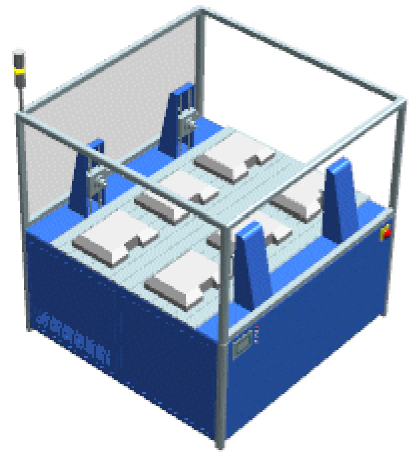




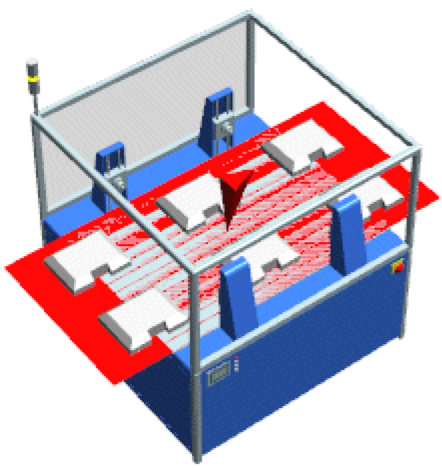
- The **ParallelStation** has a **Capacity** of four parts by default. The standard setting distributes these parts as shown below across the animation area.



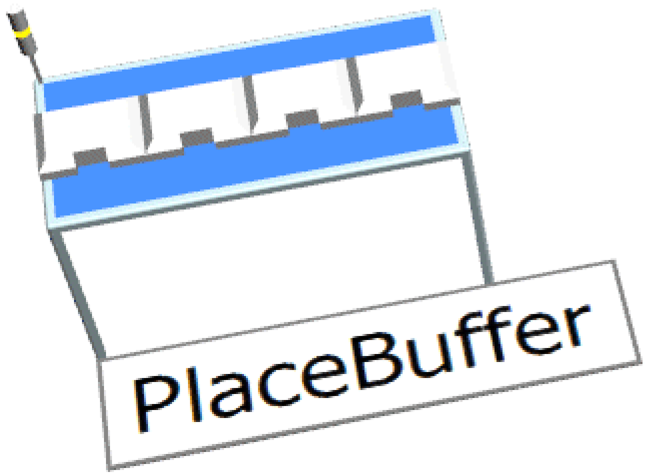
If we increase the **Capacity** in the **X-Dimension** to three parts, it looks like this.

Capacity 2 times 2	Capacity 3 times 2
	

You can also increase the length of the animation area, in our example to 3 meters. *Plant Simulation* then evenly distributes the parts across the changed animation area.



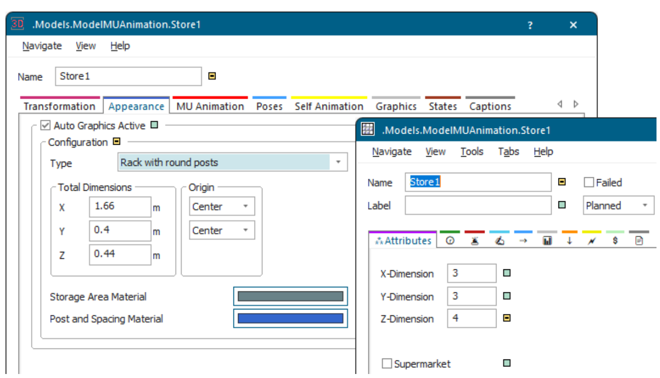
- The **PlaceBuffer** has a **Capacity** of four parts by default. It animates the parts as follows:
    - If the **Animation Area** is active, the *PlaceBuffer* uses it. Here the index of the *MU* is the index of the area in the Y-direction; the capacity in the X-direction always is 1. It looks like this for the standard capacity of four parts:

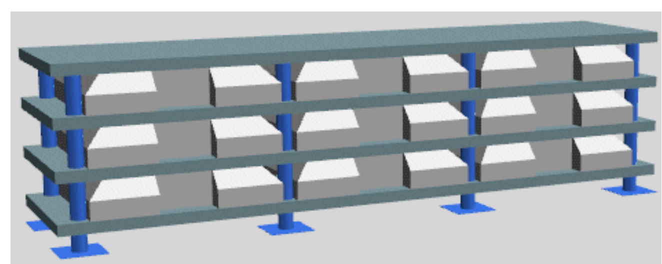


  - If a **MU animation path** named **Default** with more than one anchor point exists, the index of the *MU* is converted to a relative position. *Plant Simulation* then evenly distributes the *MUs* on this line.

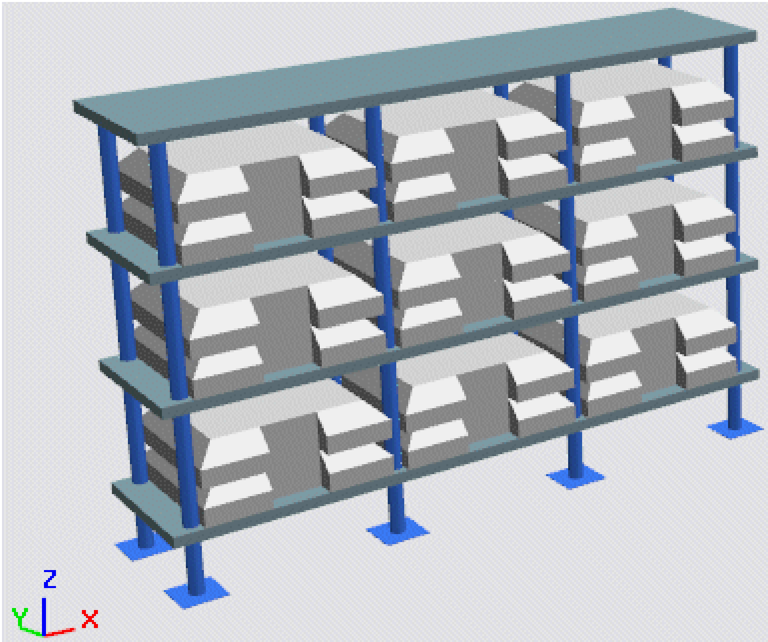
    Otherwise the *PlaceBuffer* uses the first path for animating the first *MU*, the second path for the second *MU*, etc. If no suitable path exists, but if a path named **Default** consisting of a single point exists, the *PlaceBuffer* uses that path.

- The **Store** with the standard setting, three parts each in three rows, looks as shown below with nine parts placed in stock. The parts exactly fit the individual storage compartments.

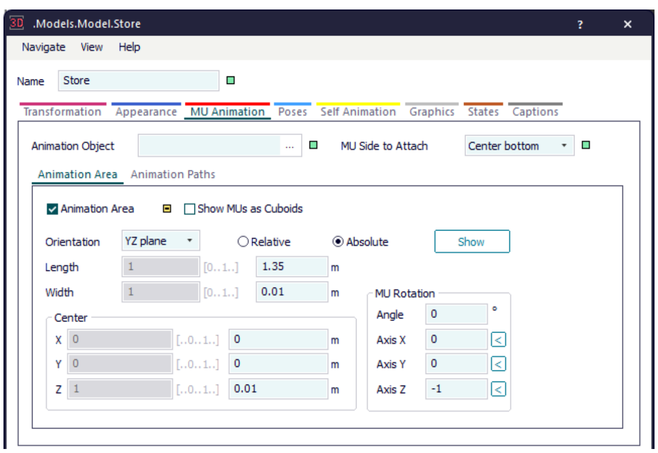


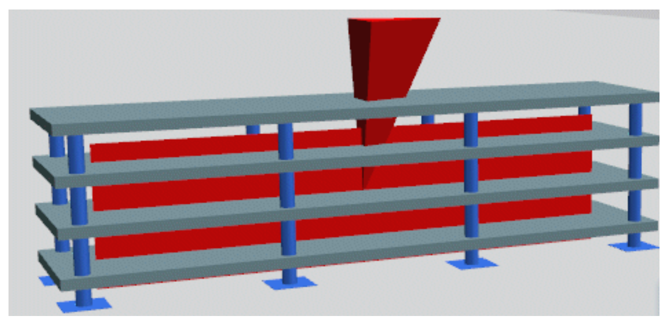


You can change the **Dimensions** of the individual storage compartments and the **Capacity** of the *Store* in the **Z-Dimension**. In the example below we increased the height in the **Z-Dimension** and the number of parts in the **Z-Dimension**. This way *Plant Simulation* stacks several parts in the individual storage compartments.

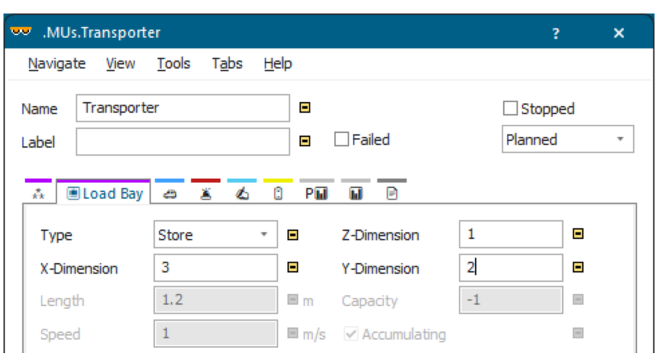


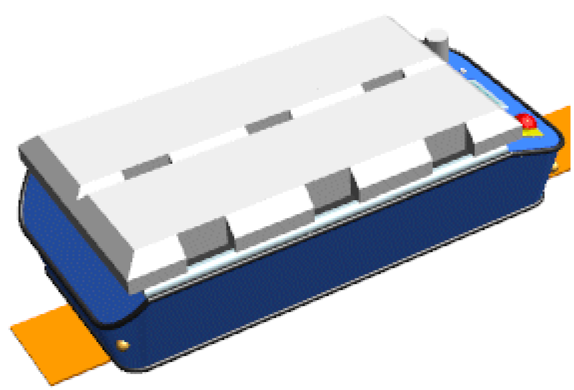
The storage area is placed on the **XZ plane** and looks like this when it is shown.



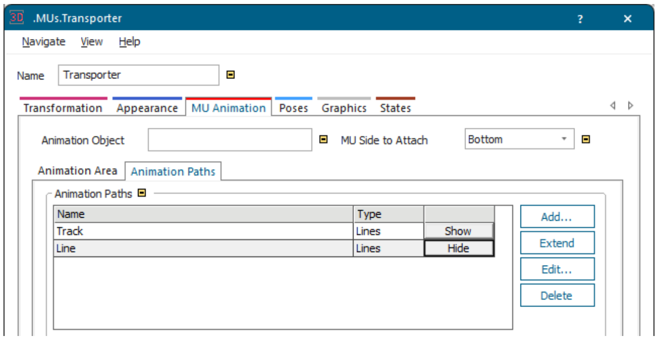


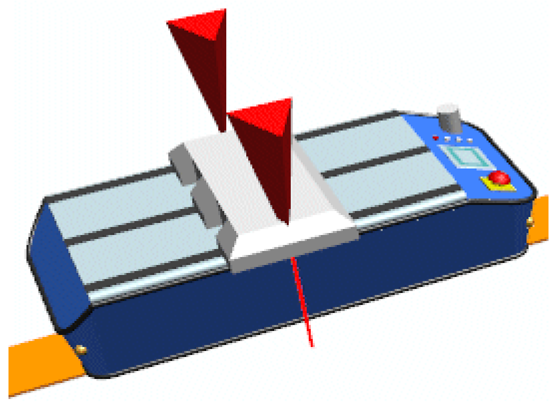
- The **Transporter** with the standard settings **Store** and **Animation Area**, with three parts each in two rows, looks like this with six loaded parts.



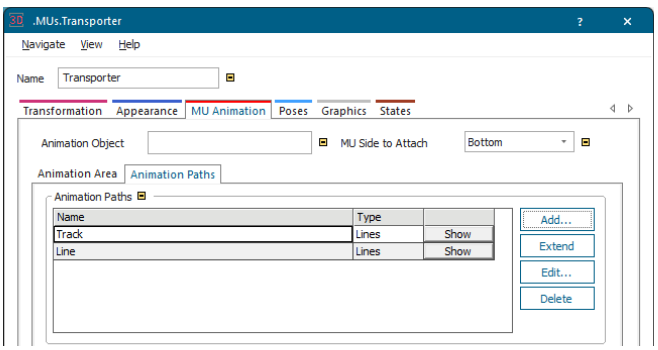


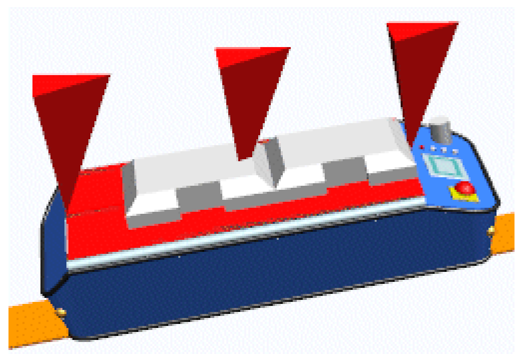
The *Transporter with an active loading space* of type **Line** uses the animation path named **Line**.



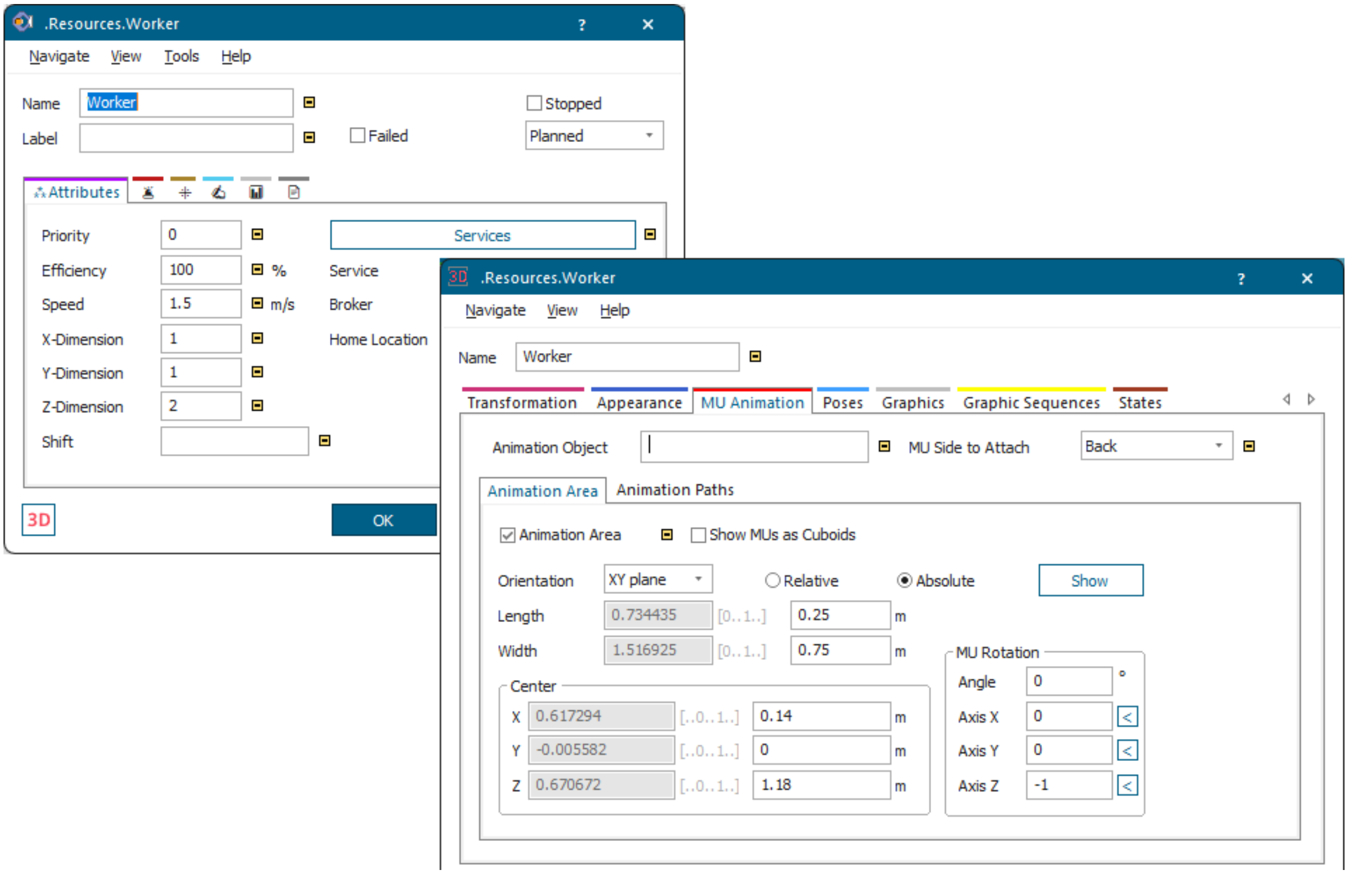


The *Transporter with a passive loading space* of type **Track** uses the animation path named **Track**.





- The **Worker** with the standard settings has a **Capacity** of one part in the **X-Dimension** and in the **Z-Dimension**.

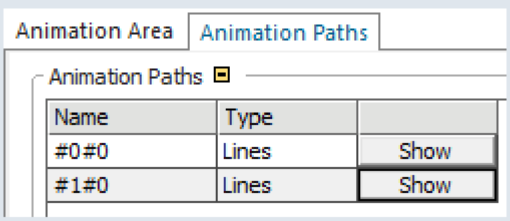


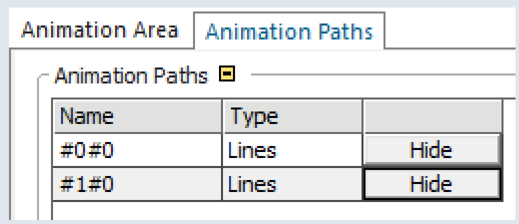
It looks like this when shown in the *Frame*.

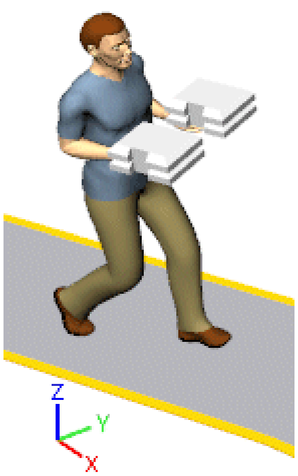
Walk without part, one part	Walk and carry part, one part
	

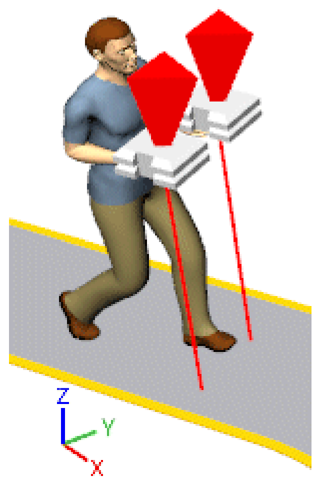
If you want the *Worker* to carry two stacked parts, type in 2 as **Z-Dimension**. This then looks like this.

If you want the *Worker* to carry two parts in each hand, type in 2 as **X-Dimension** and clear **Animation Area**. The *Worker* then uses the **Animation Paths** which we defined. This then looks like this.









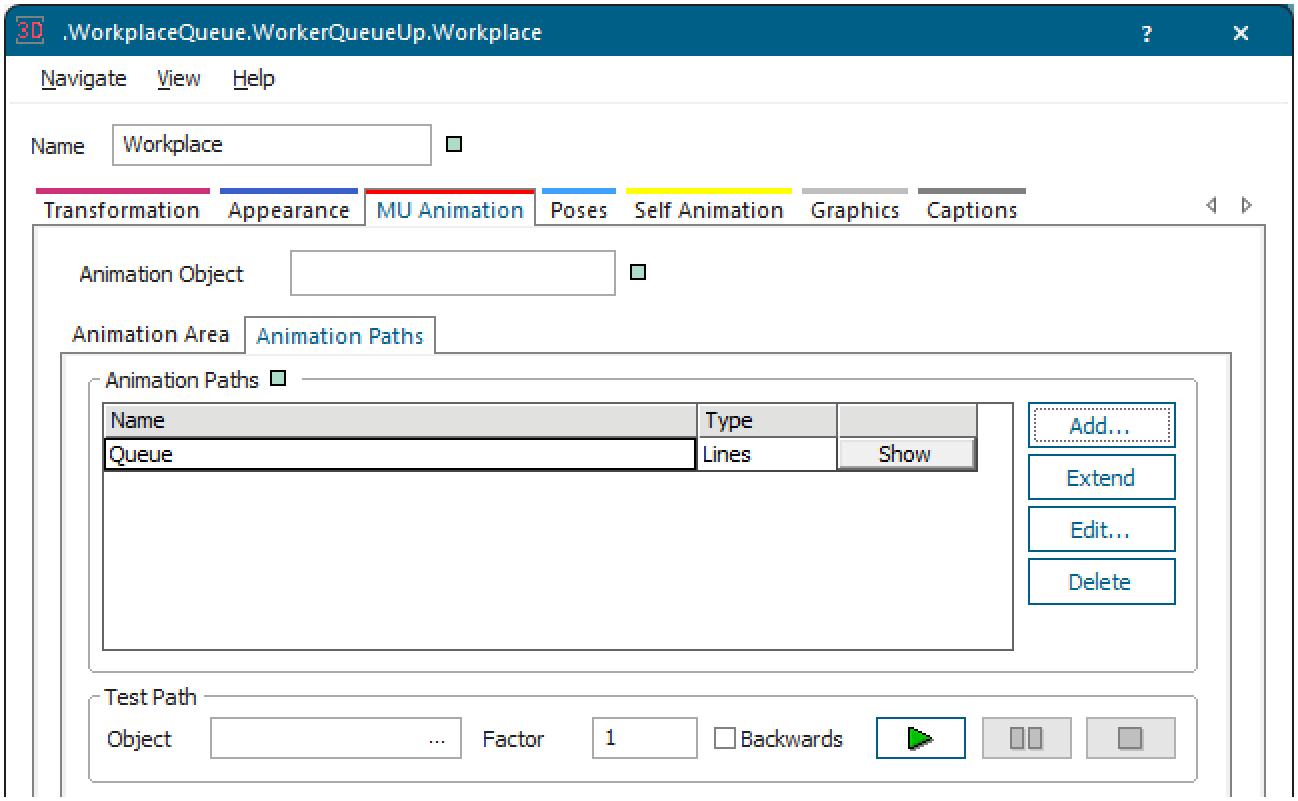
See also Model a Worker Who Carries Several Parts

Back to Control the MU Animation

MU Animation on Animation Paths on the Workplace

The *Workplace* provides a single **MU animation path** named **Queue** by default.

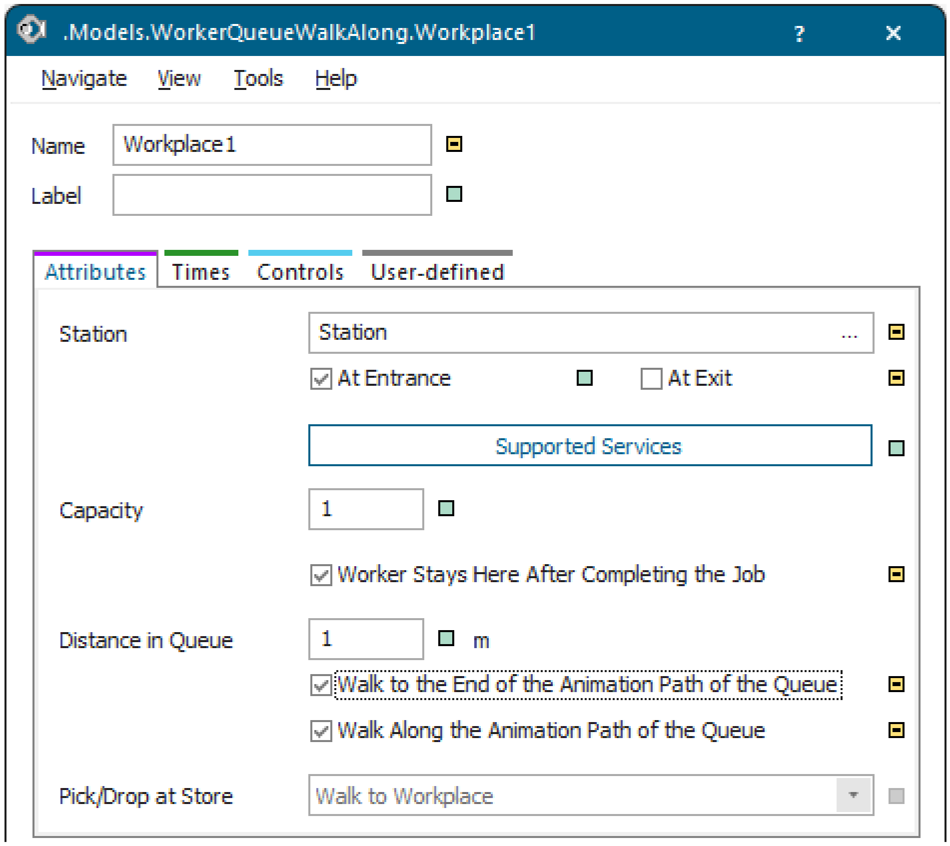
Plant Simulation animates the *Workers*, who are waiting in front a *Workplace* which they cannot enter, on this animation path that visualizes the queue.



Note

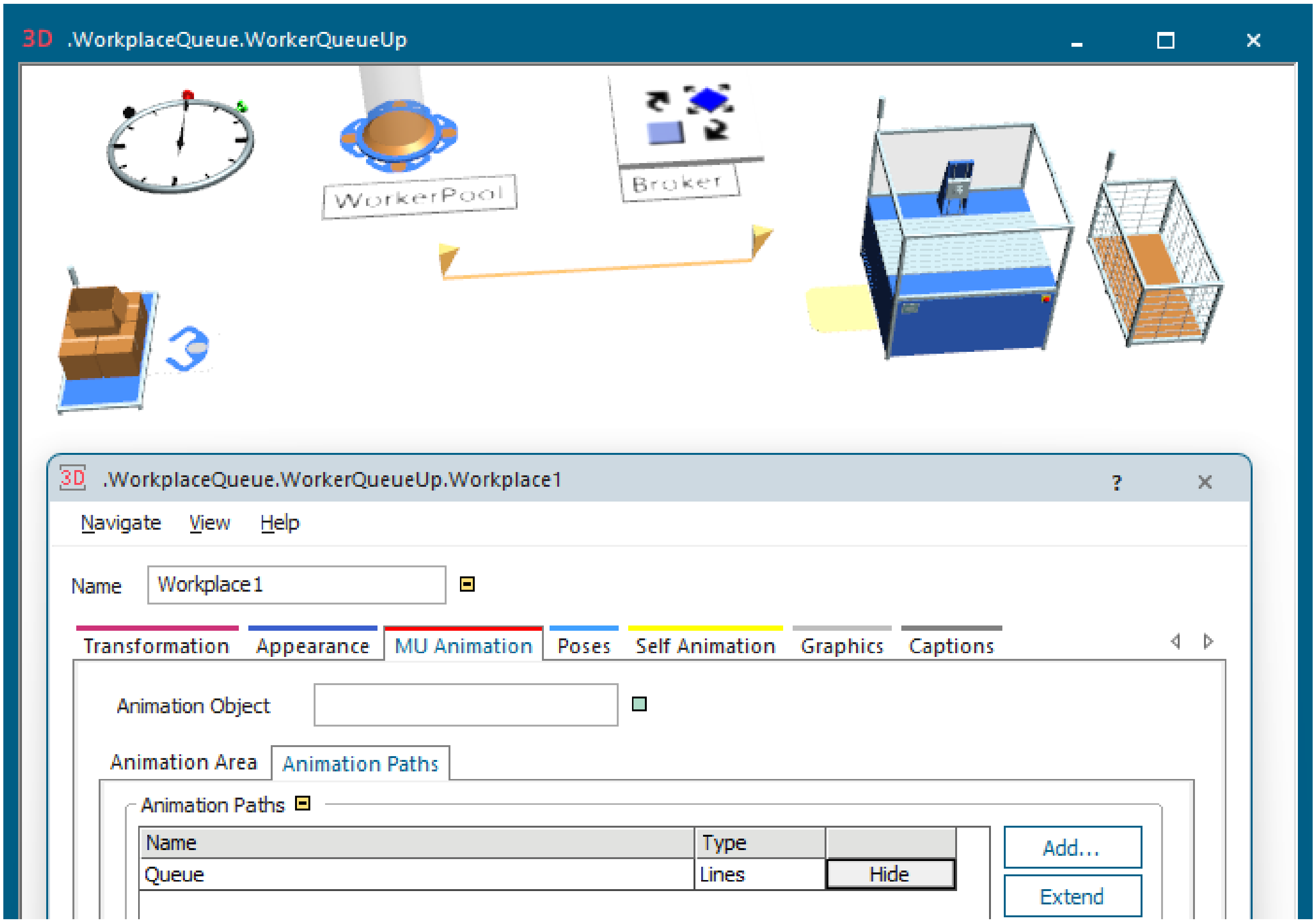
If a *Workplace* does not provide a **MU animation path** named **Queue**, all *Workers* wait at the same position until they can enter the *Workplace*.

You can set the **Distance in Queue** in the dialog of the *Workplace*.



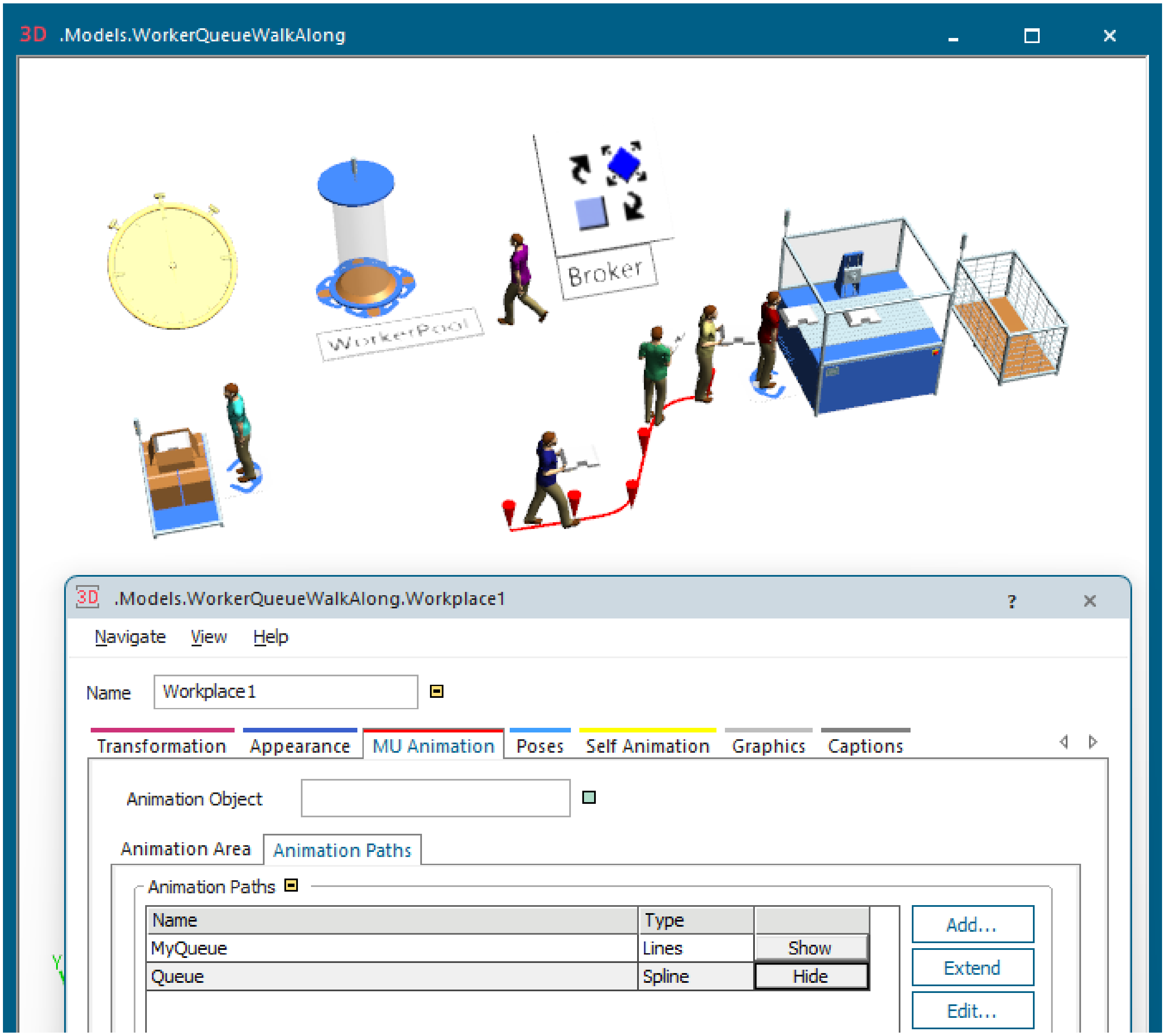
Click **Show** to show the **MU animation path**. The path tool, i. e., the pyramid, shows where the *Worker* is going to be animated. You can click the path tool and move it with the arrow keys on the keyboard. The *Worker* will then be animated at the new position.

The predefined animation path looks like this.



You can also define a path of type **Polycurve**, **Spline**, or **Rotation path**. Before you do that though, rename the original, predefined path named **Queue**.

In this example we first renamed the default path to `MyQueue` and then created a spline path named **Queue**. *Plant Simulation* then animates the *Workers* waiting in line on this path.



If you want to use the default path again, rename your path and change the name of the original path back to **Queue**.

To animate the *Worker* on an animation area, select the check box **Animation Area** and define the respective settings. Our Workplace has a **Capacity** of 3.

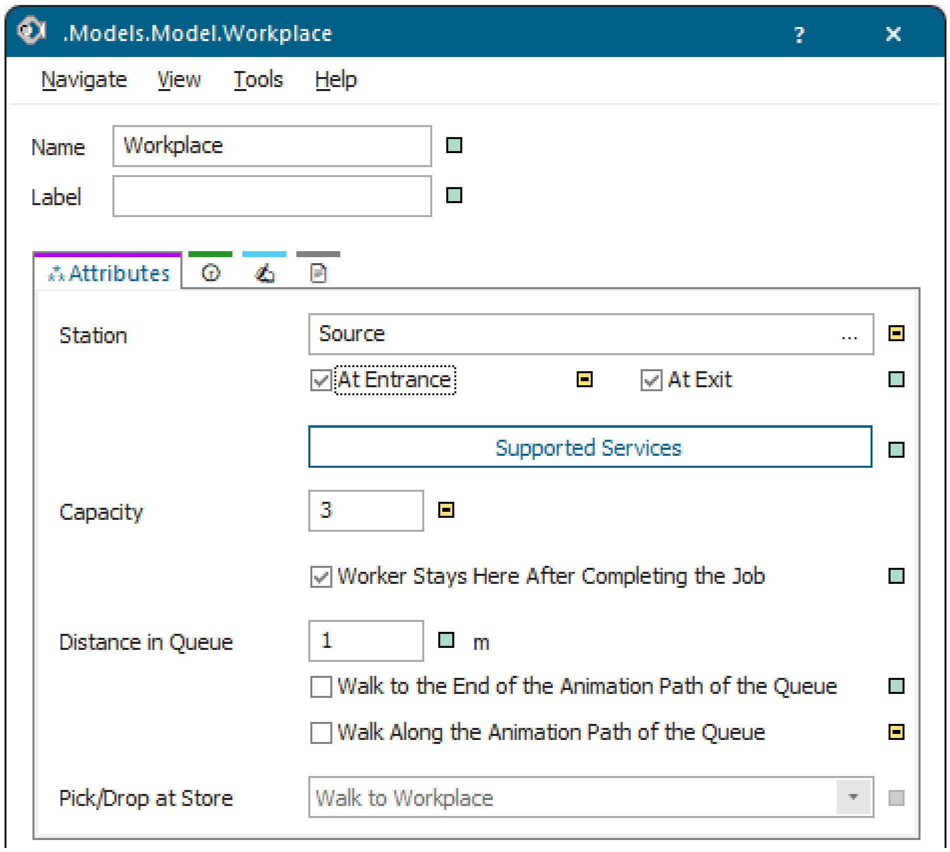
Note

If you do not use an animation area this applies:

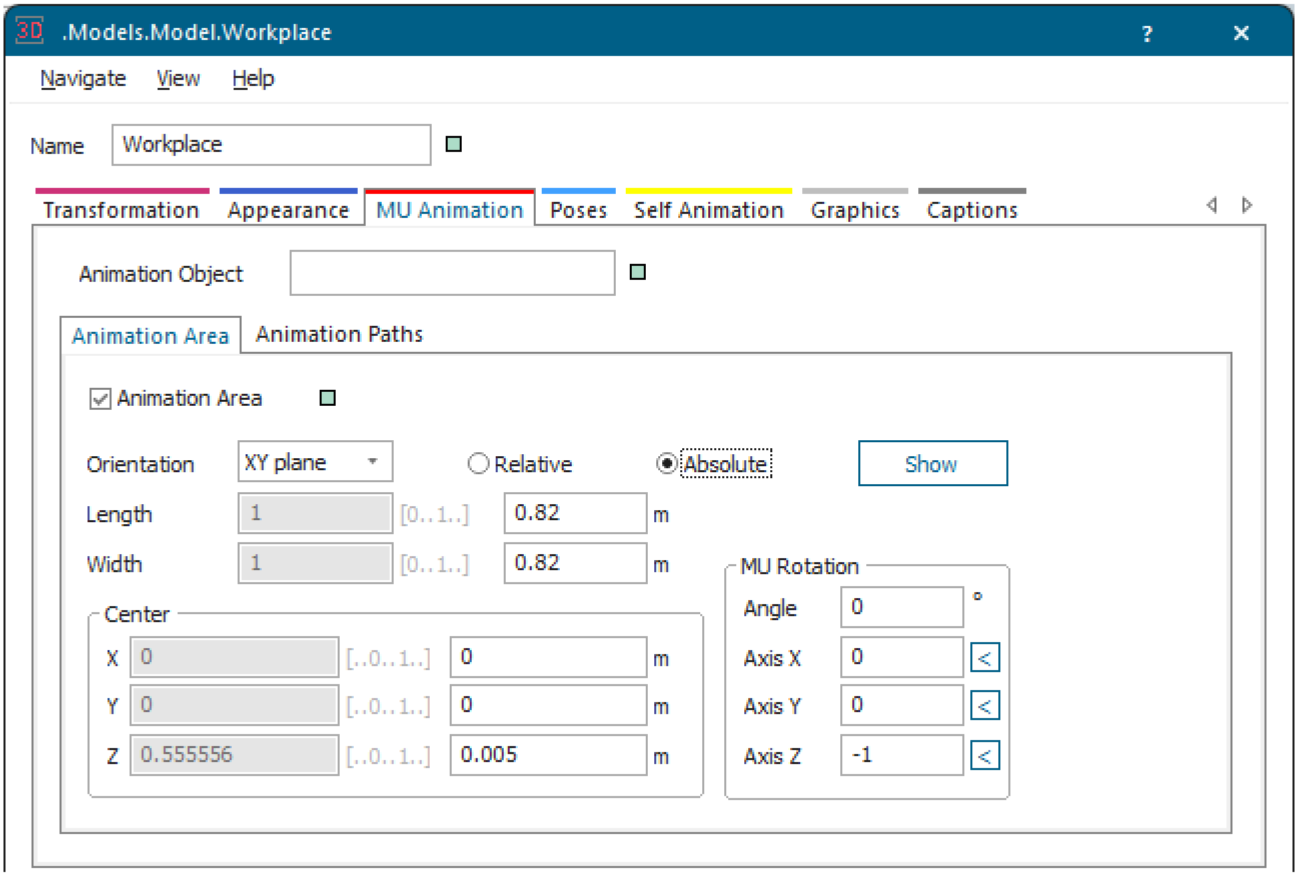
Plant Simulation animates the first *Worker* on the first animation path, the second *Worker* on the second animation path, and the third *Worker* on the third animation path. You have to define these animation paths yourself. If, for example, no third path exists for the third *Worker*, *Plant Simulation* uses the

default animation path. If no default animation path exists, *Plant Simulation* does not animate the *Worker* at all.

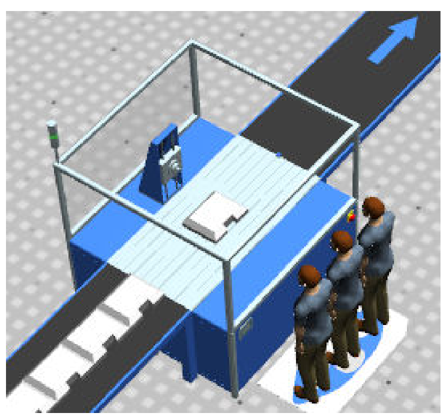
In most cases an animation path is the suitable way to animate the *Workers* on the *Workplace*. If you modeling situation requires it, you can also animate the *Workers* on an **animation area** though.



We animate it on an **Animation Area** with these settings:



This looks like this in our model:



See also

Model a Worker Who Waits in Line

How Workers Queue Up in Front of the Workplace

Back to Control the MU Animation

Video on YouTube

https://youtu.be/gU1pqu7sWA8?si=HgdhN-JsmsY7HQSH&t=775

Modeling Joints and Poses

Model Joints and Poses

In our sample model we show how to model realistic kinematics with joints and poses.

You find the respective settings on the tabs **Joint** and **Poses** in the dialog **Edit 3D Properties**.

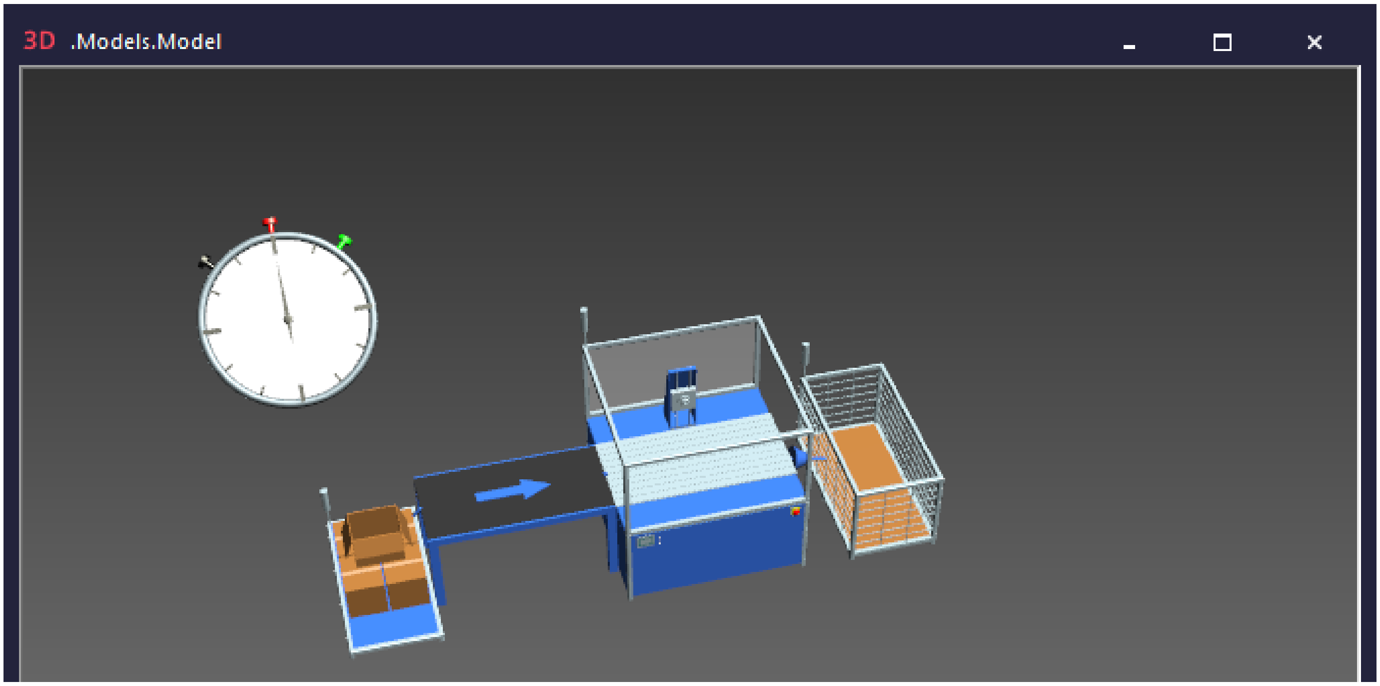
You can use poses to animate the object itself. As opposed to the normal self animation, you only have to set the place to which the animation is to move when using the animation with poses. As *Plant Simulation* knows the previous position, it then internally creates self animations which ensure a smooth transition from the previous to the new state.

In our model the *Source* produces parts. The *Conveyor* transports the part to the *processing station* which processes it. A revolute joint rotates the **ToolHead** to the right and left and a prismatic joint moves the **ToolHolder** up and down. We could also attach the part to the **ToolHolder** and manipulate it there. This would go beyond the scope of this short demonstration though. The *Station* then passes the part on to the *Drain*, which removes it from the installation.

We will demonstrate how to:

- **Configure the Simulation Objects**
- **Create the Animation Objects**
- **Configure the Joints**
- **Configure the Poses**

The finished simulation model looks like this:



The sample model **Factory 51** on the **Start Page** shows additional usages for **revolute joints** and for **prismatic joints**.

Compare Move Poses and Joints with Mouse and Keyboard Keys

Compare Create a Simulation Object with Animation

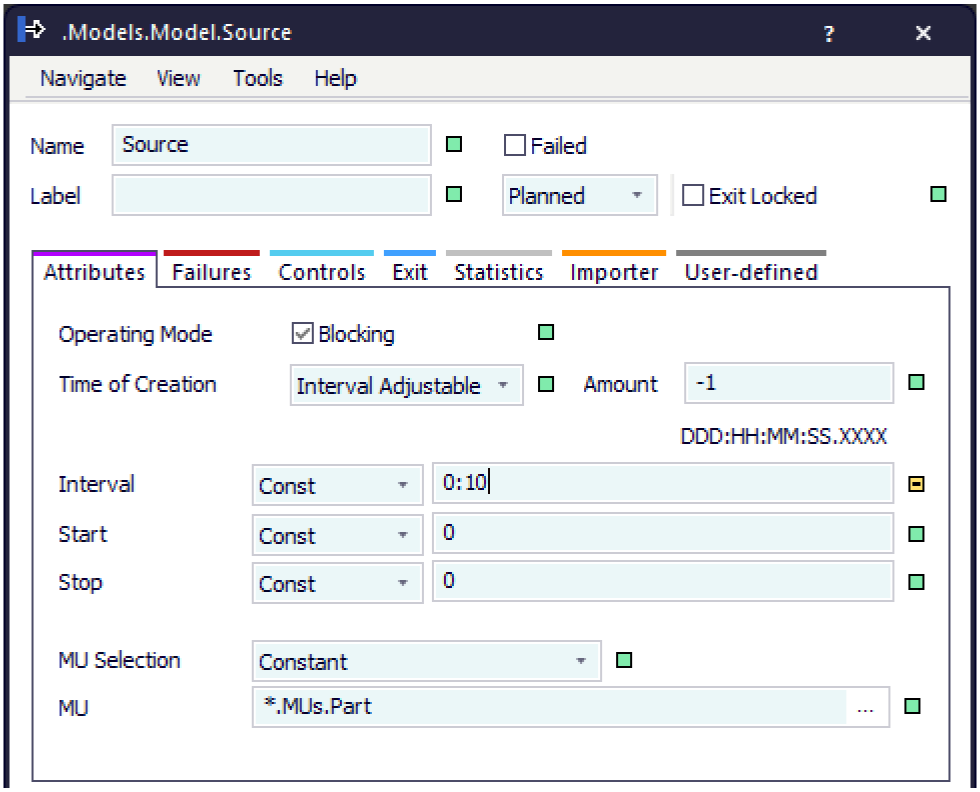
Back to Visualizing the Material Flow

Configure the Simulation Objects

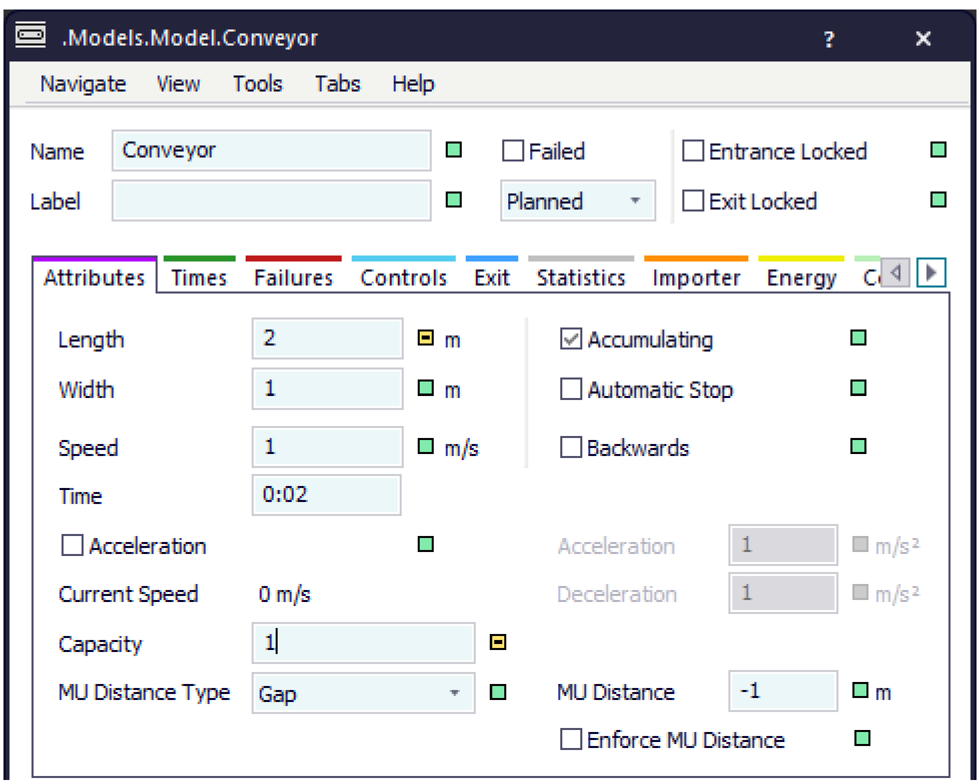
In our first step we configure the simulation settings of the objects.

Proceed as follow:

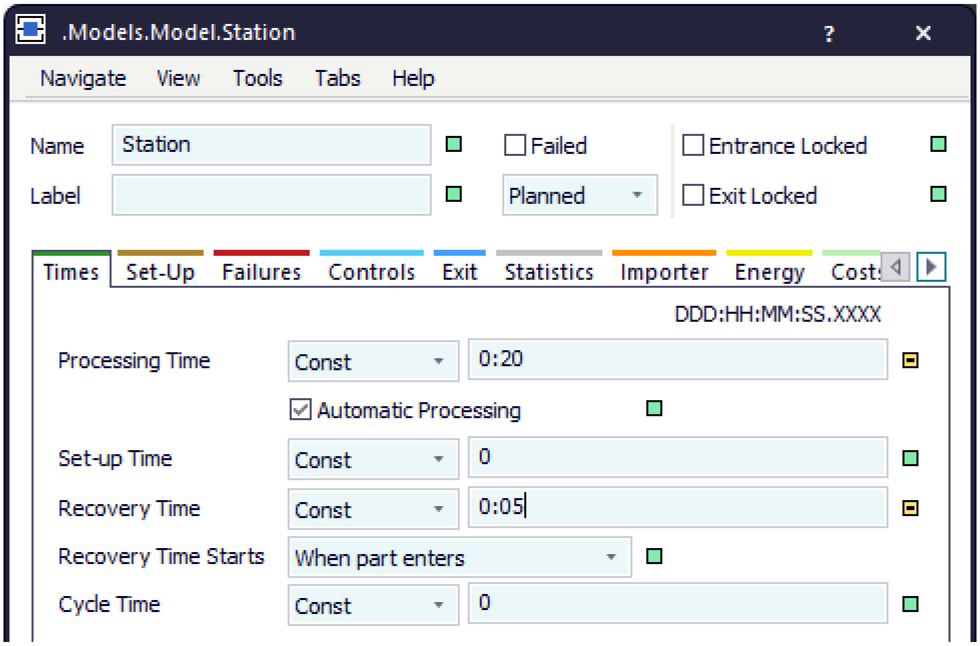
- For the *Source* we entered a constant **Time of Creation** of `10` seconds. Apart from that we use the default settings.



- For the *Conveyor* we entered a **Capacity** of 1, so that it only transports a single part. Apart from that we use the default settings.



- For the *Station* we entered a **Processing Time** of `20` seconds and a **Recovery Time** of `5` seconds. Apart from that we use the default settings.



- For the *Drain* we use the default settings.

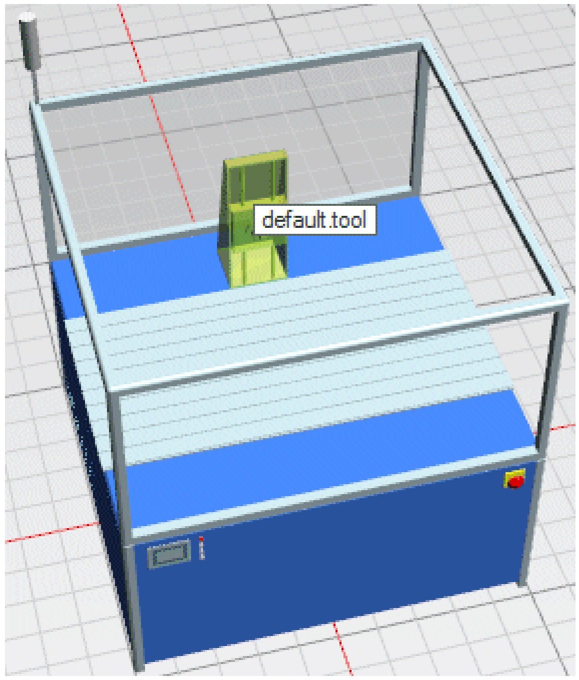
Back to Model Joints and Poses

Create the Animation Objects

When the part is located on the machine, the **ToolHead** rotates to the right and the **ToolHolder** moves up. Then **ToolHead** and **ToolHolder** remain at their new position while the **Processing Time** passes. After that they both return to their original pose.

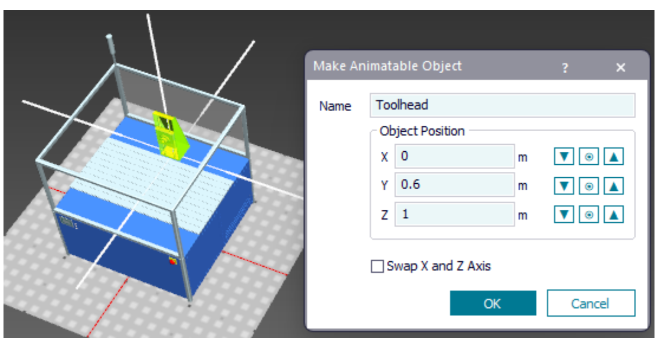
Proceed as follows to create the animation object of our machine:

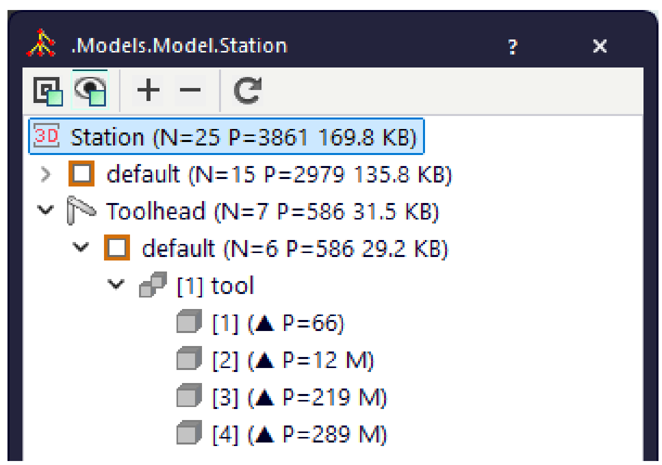
- Click the *Station* with the right mouse button and open it in a new window.
- Click the part of the graphic called **default.tool** with the right mouse button.



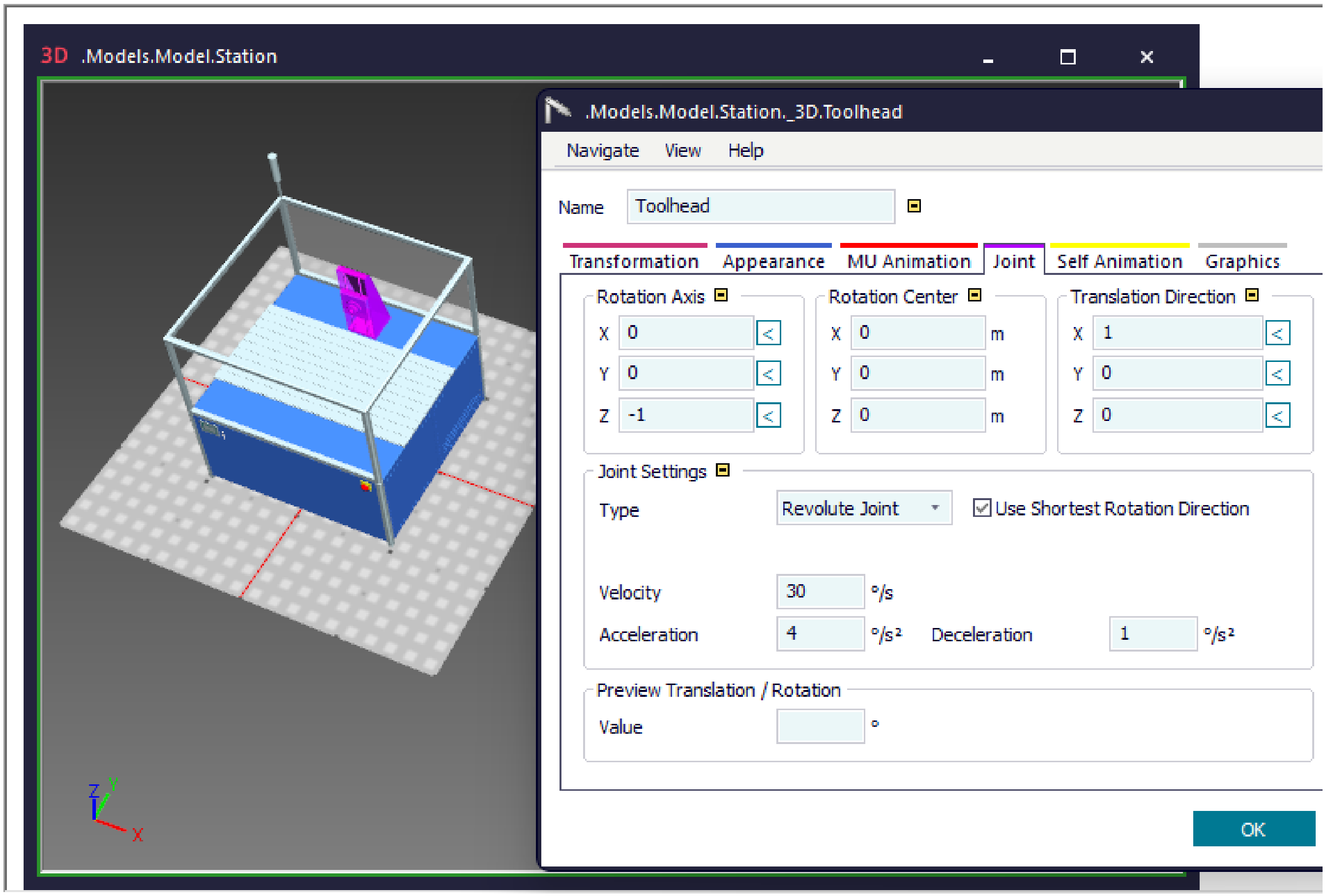
- Select **Make Animatable Object**. Type in **ToolHead** as the **Name**. Click **OK**.

> **Note**
>
> The **Object Position** sets the origin of the rotation around which the tool head will be rotated. If the object does not rotate as expected, check the values for the individual axes of the origin of the rotation.

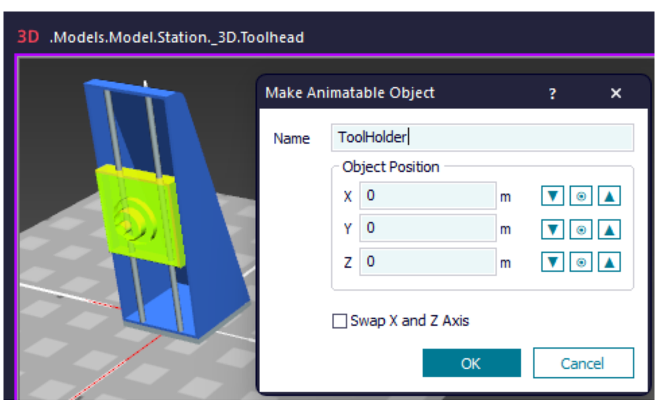


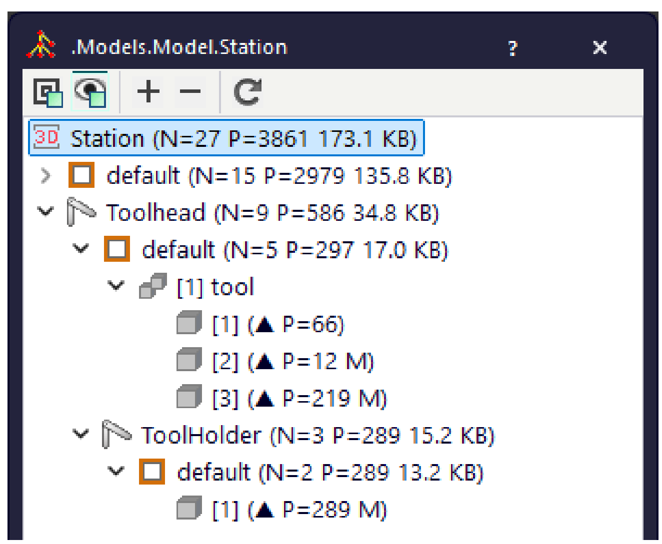


Then, set the joint settings on the **Tab Joint**. We describe this in the next topic.



- Click the **ToolHead** with the right mouse button and open it in a new window.
- Click the **ToolHead** with the left mouse button. Press the **+** key, i. e., the square part with the telescopic holder, to select the **ToolHolder**. Click it with the right mouse button and select **Make Animatable Object**. Enter as the **Name**. During the simulation the toolholder moves up and down.



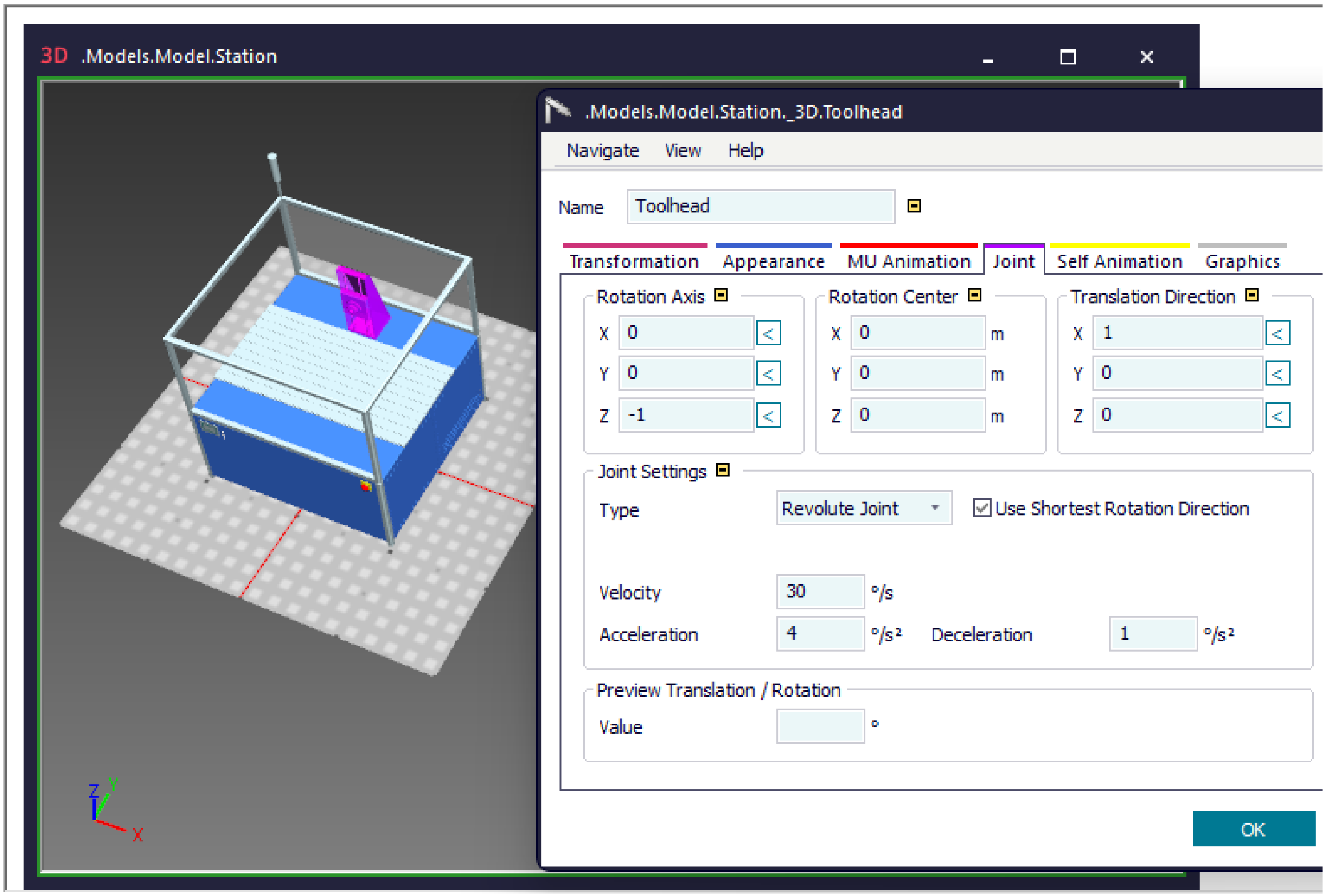


Back to Model Joints and Poses

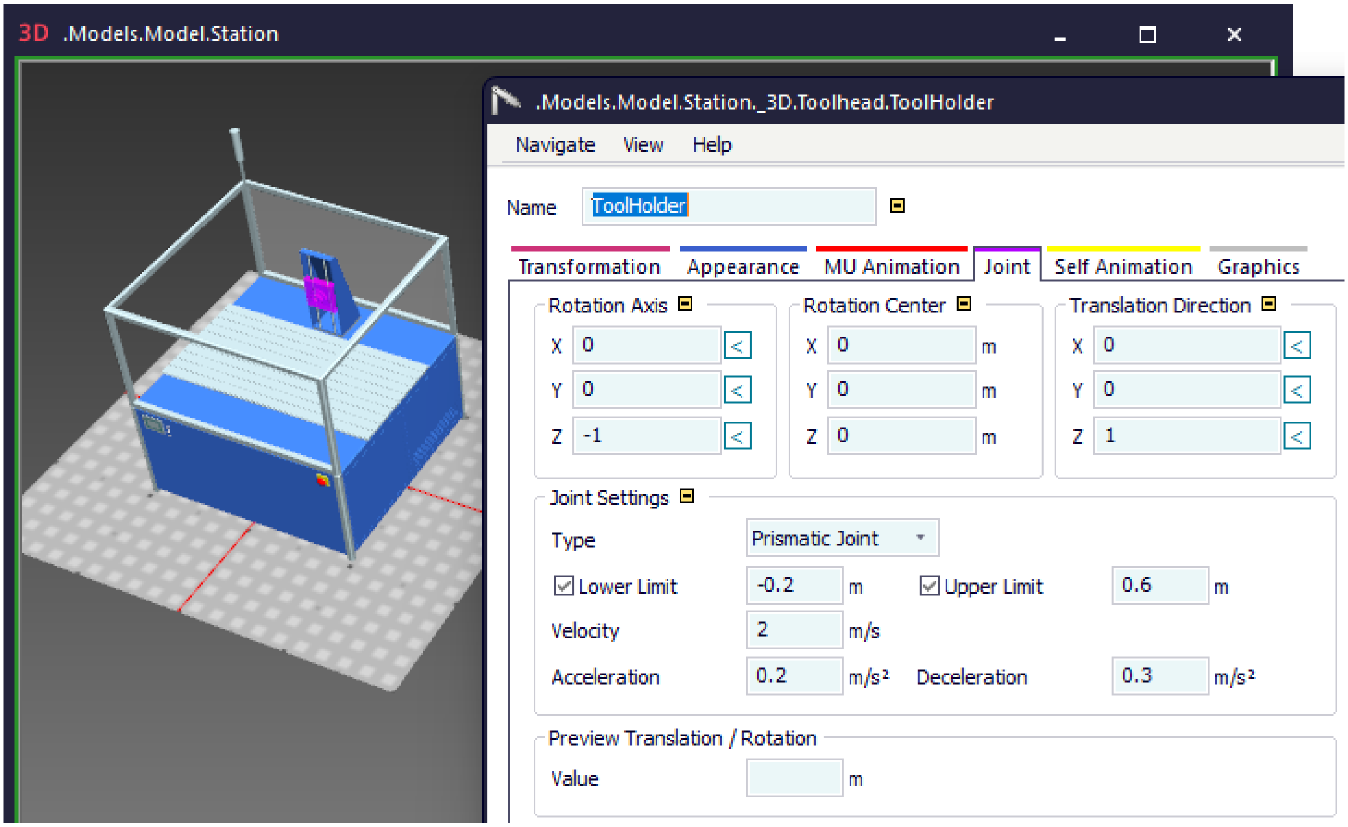
Configure the Joints

In the next step we configure the joints of the **ToolHead** and of the **ToolHolder**. We equip the **ToolHead** with a **Revolute Joint** and the **ToolHolder** with a **Prismatic Joint**.

- Click the **ToolHead** with the right mouse button and select **Edit 3D Properties**. Change to the Tab Joint. We selected the **Joint Settings** below for the **Revolute Joint**.



- Click the **ToolHolder** with the left mouse button and press the **Spacebar**. Change to the tab **Joint**. As the **ToolHolder** is to move up in the z-direction, we clicked on the button next to Z. This sets the z-direction as the **Translation Direction**. We then selected the **Joint Settings** below.



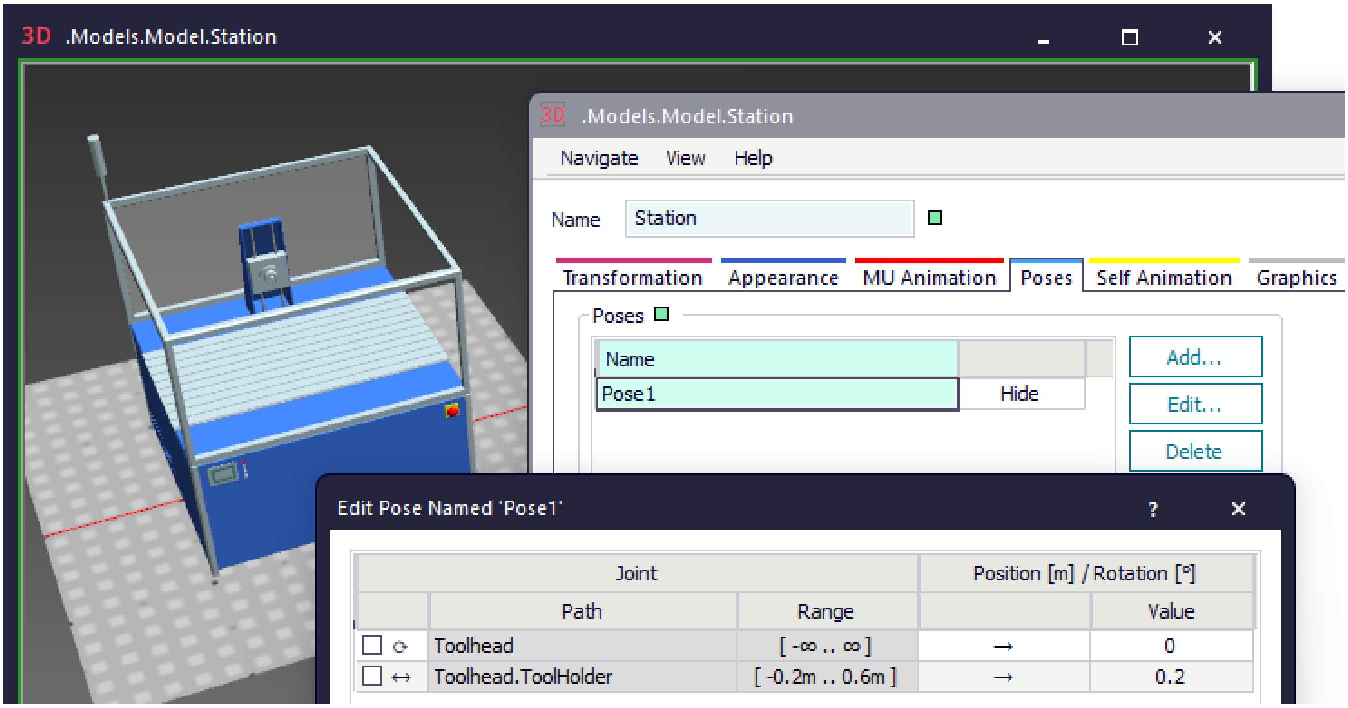
Back to Model Joints and Poses

Configure the Poses

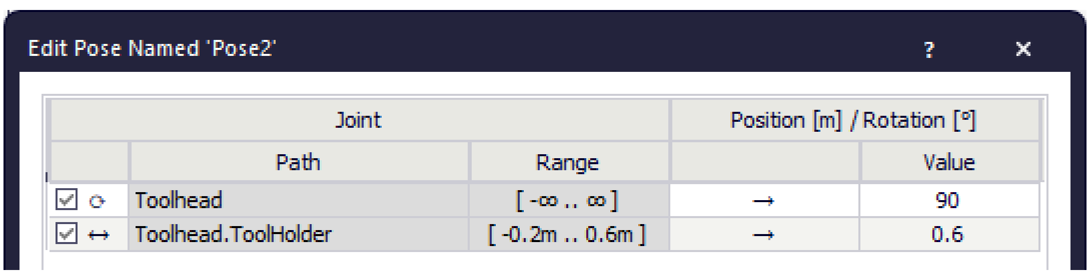
In our last step we configure the **Poses** of our *Station*.

Proceed as follows on the Tab Poses:

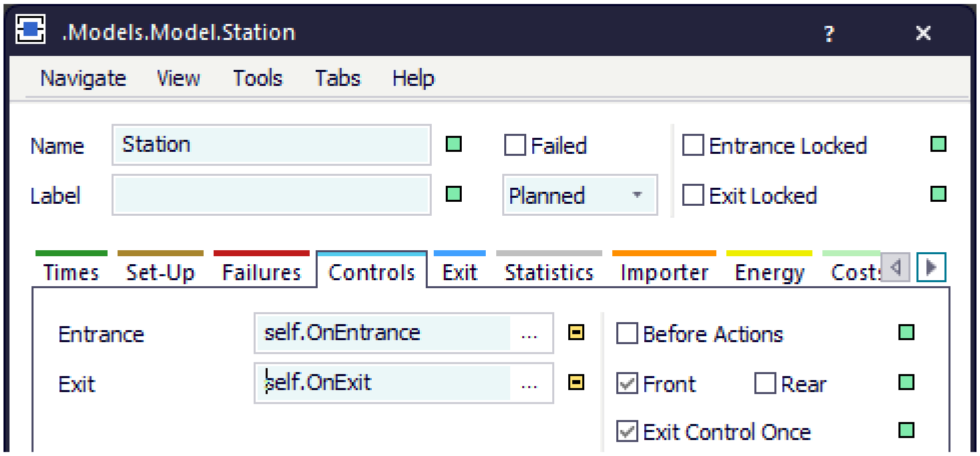
- Click into the background of the window of the *Station* with the right mouse button. Add `Pose1` and `Pose2`.



- Click **Pose1** with the left mouse button and click **Edit**. The dialog shows the settings which we selected on the tab **Joint**.
- Click **Pose2** with the left mouse button and click **Edit**. The dialog shows the settings we selected on the tab **Joint**.



- Now that we configured joints and poses, we can program the movements in the respective controls. To do so, we return to the simulation object and change to the tab **Controls**. Here we create an **Entrance Control** and an **Exit Control** each.



Within the controls we define to which poses the **ToolHead** and the **ToolHolder** move.

For the **Entrance Control** we entered these instructions:

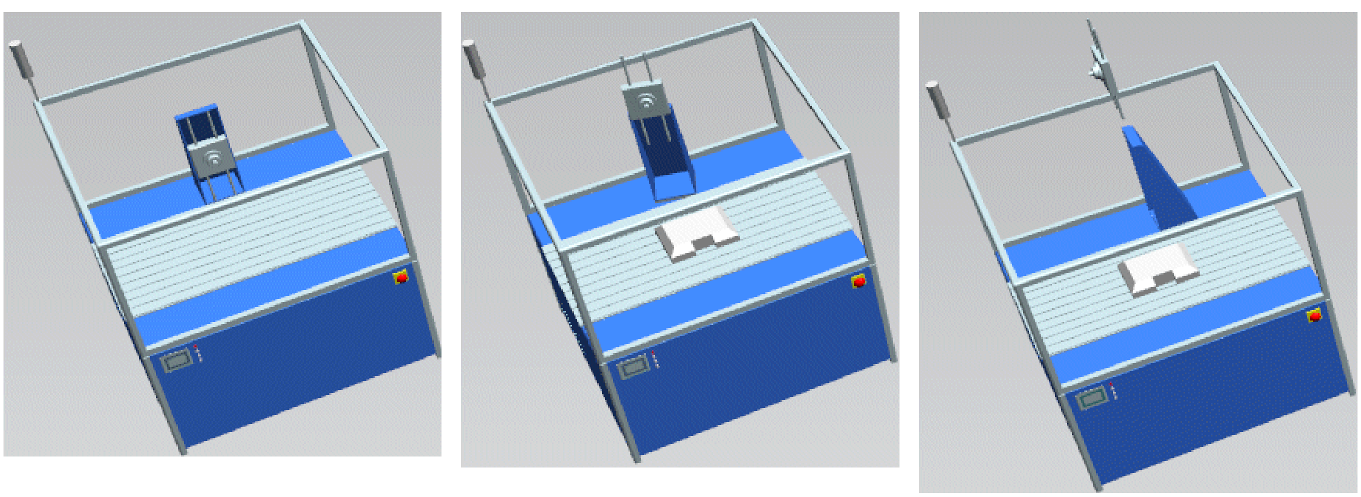
```
?._3D.Poses.moveTo("Pose2") // moves to Pose2
```

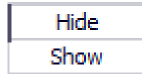
For the **Exit Control** we entered these instructions:

```
wait ?._3D.Poses.moveTo("Pose1") // returns to Pose1
@.move                           // and moves the part
```

When you now start the simulation, you can watch how the **ToolHead** rotates and how the **ToolHolder** moves up and then down again.

To better view the sequence, we selected the **Real time factor x 3**.



You can show and hide poses with the toggle button 

. *Plant Simulation* only shows a single pose at a time. If you show a different pose, *Plant Simulation* first hides the pose that is already shown.

Plant Simulation moves the object to the state that corresponds to the pose. While the pose is shown and the dialog is open, you can change the state of the pose stepwise by clicking the left or right arrow key:

Press one of the **arrow keys** to

- rotate the pose by 1 ° of the range from the joint settings if both limits are defined.
- move the pose by 0.1 m if at least one limit is missing when a **Prismatic Joint** is selected, or
- rotate the pose by 1 ° if at least one limit is missing when a **Revolute Joint** is selected.

The **left arrow key** moves the pose toward the lower limit.

The **right arrow key** moves the pose toward the upper limit, independent of rotation axis or translation direction.